

Rug Royale: Turbin3 Project Proposal & LOI

Part 1: Final Proposal · Part 2: Process Appendix (Red Team Log)

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Part 1: Final Project Proposal & LOI

Rug Royale: Turbin3 LOI, Refined (Part 1 of the submission)

Project overview

Rug Royale is a 1v1 trading-duel protocol on Solana for StonkFun traders and the pump.fun and fomo crowd they came from: people who follow KOL call-outs, place their own trades, and flex their PnL, plus retail who want a low-stakes competitive game on top of the action. Today there is no way to challenge a specific opponent, settle who is the better trader, and win money and status doing it; screenshots cannot be verified and call-outs cannot be settled. Two traders each escrow an entry, trade an identical bankroll on the StonkFun coin of their choice, both paired to the same quote asset, for a short fixed window, and the program pays the pot minus rake to the higher PnL%, with every fill and the result onchain so no referee is needed. The capstone ships the escrow, swap, and settlement program on devnet against a deterministic internal market, with a thin duel-room frontend. The same escrow-and-settle core extends to any token ecosystem, team battles, tournaments, and agent opponents.

Phase 1: Project Definition & Market Analysis

1. Core Value Proposition & PMF

Value proposition

Trading is already treated like a game by the people who do it most. As speculation goes mobile and crypto opens markets to everyone, money is following the path news took online and becoming entertainment. Rug Royale makes that game explicit and profitable: Call of Duty for trading, with intense, short, repeatable matches where you prove yourself against a real opponent in real time. Two players each escrow an entry, trade an identical bankroll on the StonkFun coin of their choice, both paired to the same quote asset, for a short fixed window, and at the buzzer positions are valued onchain and the higher PnL% takes the pot minus rake. Keeping the entry separate from the bankroll bounds each player's loss and keeps PnL% comparable. Because escrow, fills, and settlement all happen onchain, results cannot be faked and settlement needs no operator; in Web2 the same product needs licensed custody before the first match. Focusing on StonkFun coins creates real incentives around each paired community and turns duel activity into drip for participants, a multi-sided model where players, coin communities, and the launchpad all gain from every match. The protocol earns a rake on each pot and a fee on each swap. Our tagline: "Stop flexing fake screenshots. Settle it onchain."

Why now

1. Social trading is a measured business, not a trend. fomo's weekly revenue crossed \$2M and it briefly overtook Axiom in daily fees on 2026-08-06 [S1]. pump.fun answered with

- its own social trading feature the next day [S1]. Streamers on both need new interactive segments, and duels let them accept challenges from each other or from their audience.
2. Competitive trading demand is proven and the head-to-head, stake-your-own format is not served. fomo runs sponsored many-player PnL competitions [S2]; nobody runs 1v1 matches with escrowed stakes on real fills.
 3. StonkFun is the moment. After its Raydium LaunchLab integration around 2026-09-06, STONK reached roughly \$140M market cap on roughly \$135M daily volume [S3], and the ecosystem's reward-mode coins pay holders in the quote asset, which gives duels a built-in drip [S4].

Later economics (see Go-to-market and growth): a Rug Royale coin launched on StonkFun in reward mode and paired with STONK would let the rake buy the coin and drip STONK to duelists; post-capstone option pending counsel.

Product-market fit hypothesis

First retained cohort. StonkFun traders who already call each other out over a launch in the StonkFun and X communities. They have a coin they care about, an opponent they want to beat, and a habit of posting the result. The duel gives the call-out a referee and a pot.

Wedge. Freeroll and friendly duels on the week's momentum StonkFun coins, promoted inside those coin communities and, if StonkFun agrees, featured by the launchpad because every duel pushes swap volume through its pools and burns STONK.

Retention hook. The result card: two PnL curves, the transaction links, the opponent's handle, posted where the call-out started. A rematch button on the card. Reward-mode drip earned during the duel shows on the card as a second reason the match paid.

What PMF looks like, in order of proof.

1. Freeroll duels complete at a high rate and produce rematches without prompting.
2. Players return for a second session within a week without a sponsored prize.
3. Friendly duels with real entries appear organically between players who know each other.
4. A StonkFun coin community runs a duel night on its own initiative.

Revenue is not a PMF signal at this stage; retention on freerolls is.

Evidence today. Behavioral: PnL flexing and call-outs are constant in pump.fun, fomo, and StonkFun chats. Business: fomo revenue and StonkFun volume show money already flows to social and competitive trading products. Product: none yet; no duel has been played.

Top unvalidated assumptions. (1) Skill shows up at short windows; if a ten-minute duel is a coin flip, the better trader does not win reliably and neither retention nor the skill-contest position holds. (2) Two players can be matched at the same moment on the same coin; matchmaking liquidity kills most PvP products. (3) StonkFun attention persists long enough to seed the first cohort; if it rotates, the same product runs on whatever coins are hot, but the first community must be rebuilt.

Game modes (definitions)

Mode	Entry	Pot	Ranked	Purpose	Capstone scope
Unranked	None	None; points and a result card only	No	Practice, first-time players, content. The on-ramp.	Yes
Freeroll	None	Sponsored by a coin community, StonkFun, or a streamer	No	Community duel nights, launch events	Yes
Friendly	Both players opt in	Both entries, winner takes the pot minus rake	No	Two players who know each other settle a call-out	Yes, devnet only
Ranked	Required	Both entries minus rake	Yes: random matching, seasons, the Ladder	Competitive play	No. Post-capstone, after counsel and geo-blocking

Notes. Unranked and freeroll are the same on-chain duel with `entry_lamports = 0`; a freeroll is one where `sponsor_prize` funded the pot before the duel started. Points are off-chain (indexer and display), consistent with the state-ownership rule; the on-chain record is the settled `Duel` account. A mode with no entry removes “consideration” from the prize-chance-consideration test, which is why unranked and freeroll are the launch modes.

Coverage check: value drivers

Driver	Covered	Where
Prove you are better against a specific opponent, in real time	Yes	Core; leads the value prop
Win money with bounded loss (entry separate from bankroll)	Yes	Core
Short, repeatable matches (Call of Duty shape)	Yes	Core
Results cannot be faked; settlement needs no operator	Yes	Enabler
Community incentive (StonkFun coin communities, drip to participants)	Yes	Multi-sided model
Launchpad incentive (swap volume, STONK burn, rake destination)	Yes	GTM / partnership
Shareable content (result card, rematch)	Yes	Retention hook
Streamer segment (accept challenges from peers or audience)	Yes	Why now #1 / GTM
Skill rating over time (Ladder seasons)	Later	Phase 6
Spectator markets (Melee)	Later	Growth section, prospective venue
Learning value (replay the winner’s fills)	Not covered	Candidate driver; cheap to add to the result card
Sponsor-funded freerolls (coin teams fund prizes)	Not covered	Candidate driver; fits the StonkFun community model
Agent opponents	Later	Phase 2

2. Target Markets & User Profiles

Duel format. Each player picks a StonkFun coin; both coins share one quote asset (STONK at launch). PnL% is measured in the shared quote, so rival-coin duels compare fairly. Two holders of the same coin can duel on it as the special case. The launch narrative is “my coin versus yours.”

Target market segments (5)

1. **StonkFun call-out traders.** Holders and traders of StonkFun coins who already argue about which launch is better and post PnL. They want to settle it against a specific rival for money and status. Primary players and the first retained cohort.
2. **StonkFun coin creators and launch teams.** They want buzz and volume for their coin. A rival-coin duel night is cheaper promotion than a developer buy, and every match routes swaps through their pool and drip to their holders. Freeroll funders and promoters.
3. **fomo users and the wider memecoin origin pool.** fomo’s social traders first: they already copy-trade, follow leaderboards, and enter PnL competitions, so they are the likeliest early adopters outside StonkFun. Behind them, daily traders on pump.fun, Axiom, Photon, and BullX who post PnL. This is where StonkFun traders came from and the expansion pool once the format is proven.
4. **Streamers, later.** pump.fun and X trench streamers who could accept challenges from peers or their audience as a segment. Gated on PMF; content first, onchain second.
5. **Spectators, later.** Trench-stream viewers who want a rooting interest. Depends on a prediction-market venue such as Melee existing and staying permissionless. Growth section.

Later: trading-agent builders (Solana Agent Kit, elizaOS, Colosseum teams) as opponents and entrants, Phase 2.

User profiles (top 3)

Profile	Who	Job to be done	Today’s workaround	Why Rug Royale
The StonkFun call-out trader	Holds one or more StonkFun coins, lives in the coin’s chat and on X, argues about which launch is better, posts PnL	Beat a specific rival on the record, and win money and status doing it	Screenshots, chat call-outs, unverifiable side bets	A ten-minute rival-coin match with escrowed stakes, onchain result, and a result card to post where the argument started
The coin creator	Launched a StonkFun coin, wants volume and holders, already uses dev buys, airdrops, and reward-mode drip to promote	Build buzz and trading activity for my coin	Dev buys, airdrops, paid skills, giveaways	Fund a freeroll or host a rival-coin duel night; every match pushes swaps through my pool and drip to my holders
The fomo social trader	Trades memecoins daily on fomo, copy-trades top wallets, watches the leaderboard, enters PnL competitions, flexes PnL	Prove I am the better trader against a specific person, not a leaderboard	Screenshots, fomo competitions, unverifiable bets between friends	A 1v1 with a real opponent and escrowed stakes, on the same social rails they already use; the format follows the crowd once proven on StonkFun

Demand coverage check (measured bets)

Every row is an assumption stated openly. “Test” is the cheapest way to turn it into evidence; “Run” records whether it was done before submission.

Segment	Evidence today	Assumption	Cheapest test	Run
StonkFun call-out traders	Call-outs and PnL posts are constant in StonkFun and X chats. Ecosystem scale: ~\$135M daily volume after the LaunchLab integration [S3], ~55K STONK holders, ~1.3K launched coins [S5].	A call-out converts to a duel.	Five manual rival-coin duels with friends, escrow by hand, one afternoon.	No
StonkFun coin creators	Creators already spend to promote: dev buys up to 50% of supply, holder airdrops, reward-mode drip are built into the launch tooling [S6].	A creator will fund a freeroll or host a duel night.	DM three creators of this week’s momentum coins with one line.	No
fomo users and origin pool	fomo weekly revenue > \$2M, 625,000 registered traders adding ~3,500/day [S1][S13], and pump.fun’s social trading launch show money flows to social trading; fomo PnL competitions have entrants [S2].	They will follow into a 1v1 format, and eventually pay entry.	Post a mock result card on X; count replies and DMs.	No
Streamers, later	Streamer bounties and creator fees show segments are monetized.	A duel segment holds viewers.	One freeroll on one small stream.	Not before PMF
Spectators, later	Pumpcade and Polymarket markets on stream moments exist.	A venue exists and resolves against the Duel account.	Create one test market on Melee for a mock duel.	Not before PMF

These products are permissionless and the segments have not been interviewed, so this LOI makes a measured bet and states it. The StonkFun team and coin creators are reachable on X and will be contacted before mainnet.

Go-to-market and growth (StonkFun-first)

Launch order: unranked duels inside two or three StonkFun coin communities with momentum, then sponsored freeroll duel nights hosted by coin creators, then friendly duels between players who already know each other. The pitch to StonkFun is direct: every duel routes swap volume through its pools and burns STONK, and the protocol rake can be pointed at a STONK burn. Success in the first month is one coin community running a duel night on its own.

Later channels, each gated on that signal: - Streamers accepting challenges from peers or their audience. - A spectator “who wins?” market on a permissionless venue such as Melee, with native markets a v3 decision after counsel. - A Rug Royale coin launched on StonkFun in reward mode so the rake drips STONK to duelists. - N-player tournaments and agent entrants on the same escrow.

Roadmap after PMF. Each item ships only after the PMF signals above, on the same escrow-and-settle core, in this order: 1. **Ladder**: ranked seasons with random matching, after counsel and geo-blocking. 2. **1vN battle royale**: N players, one window, one bankroll each, top PnL% takes the pot, live elimination line. 3. **Team battles**: squads of two to four, pooled PnL%,

community versus community. 4. **Agent face-offs**: agents enter the same duels via x402, on human, agent, and mixed ladders. 5. **More markets**: any Solana token ecosystem behind StonkFun; xStocks-quoted duels only after counsel and geo-gating. 6. **Audience markets**: spectator “who wins?” markets on a permissionless venue first; native markets a v3 decision after counsel.

3. Competitor Landscape

Primary research. We searched DefiLlama raises, Colosseum hackathon projects, GitHub, and X in the original research (August 2026). An AI coverage check re-ran DefiLlama raises, GitHub (pump.fun and StonkFun API repos), and X-indexed sources on 2026-09-11, plus direct fetches of product sites. Colosseum result: the only related project found was Pregame, a sports-betting competition product; no trading-duel project surfaced.

The niche is open. No StonkFun-native or third-party product runs duels, competitions, or leaderboards on StonkFun coins as of 2026-09-11. The only third-party tool found is The Stonk Board, a dashboard of top reward coins by APR and holder payouts [S7]. The StonkFun developer API exposes no competition, leaderboard, or PnL features [S6].

Product	Type	Model	Traction signal	Gap vs Rug Royale
SolDuel	Direct	Pick a champion token, best % change wins; onchain escrow; mainnet, audited	0 volume, 0 matches, 0 traders [S8]	The closest model to a buy-and-hold rival-coin duel, so the difference must be explicit: Rug Royale keeps trading open for the whole window so timing and skill decide it, runs short windows, launches unranked and freeroll, and centers each duel on two coin communities whose holders earn drip from the volume. SolDuel had none of that. Two readings of its zero: escrow alone is not a product, or the category has no demand. The validation sprint separates them.
TradeFighters	Direct	Real-time trading duels between traders with a leaderboard; general Solana trading, no ecosystem focus	Product exists (checked 2026-09-11); no published traction [S9]	Broadly similar concept. No sharpening around a specific ecosystem or community, no coin-community rivalry or drip, and no stated roadmap beyond duels. Rug Royale’s differentiation is the StonkFun wedge, the rival-coin format, and the granularity of its bets (modes, tiers, windows as config).

Product	Type	Model	Traction signal	Gap vs Rug Royale
PerpClash	Direct, different market	Timed 1v1 duels on BTC, ETH, and SOL perps, up to 20x, wallet sign-in, no deposit; season points feed a prize vault split among the top 100	Does not appear live (checked 2026-09-11); site returned 404 on fetch [S10]	Perps on majors only, no escrowed stake per duel, no coin communities. Competes for the “trading duel” mindshare on Solana if it ships. Its season model is a reference for the Ladder feature.
PvPBattle	Direct	Virtual positions settled against Binance mark price; custodial SOL deposits; 30-second to 15-minute battles; up to 25x	Live; example pots 0.5 to 4 SOL; no published traction [S11]	Paper fills and custody. Nothing settles onchain, no coin communities, no drip. Its 30-second battles show the market skews fast, which sharpens the skill-at-short-horizon question.
fomo competitions / FomoVille	Substitute, fast-follow	Sponsored many-player PnL competitions, leaderboards, prize city; social and copy trading app	\$75M Series B June 2026 after \$17M Series A; weekly revenue above \$2M; third in Solana 7-day protocol revenue [S1] [S12]; \$75M Series B at a \$550M valuation led by Index Ventures with USV, 625,000 registered traders adding ~3,500/day at the time [S13]	No stakes, no head-to-head. A \$75M company already running PnL competitions can ship 1v1 in a sprint; speed is not a moat. What is defensible: the StonkFun community focus, being first in an unoccupied niche, and a permissionless protocol any frontend can embed.
pump.fun social trading	Substitute, fast-follow	Launched a social trading feature on 2026-08-07 in response to fomo [S1]	Reported new ATH in daily active users after launch [S1]	Same fast-follow risk as fomo, with the largest launchpad audience. Same defensible differentiation.
Duel Duck	Substitute	Yes/no prediction “duels” collateralized in any SPL token; creator-set rules; admin or self resolution; contract payout	Live [S14]	Betting on an outcome, not competing on your own trades. Owns part of the word “duel” on Solana; naming and messaging must separate the two.

Product	Type	Model	Traction signal	Gap vs Rug Royale
Kairos TradeX	Adjacent	Trading terminal for Polymarket and Kalshi; ran a trading championship with \$30K across Net PnL, ROI, and Volume pools	\$2.5M seed Feb 2026, a16z crypto and Geneva Trading [S15]	Sponsored many-player tournaments on prediction markets. Not head-to-head, not memecoins.
Big Dance	Substitute	Tournament prediction market: buy tokens on a team, loser pools merge into the winner's; 100% onchain; 50% referral fee share	Live for March Madness 2026 [S16]	Event bracket, not trading skill. Its referral-share model is a GTM idea worth borrowing.
DUELPVP / LootScape	Adjacent	Dice, coinflip, VRF for SOL	Live	Chance, not skill; no content angle.
TradeFlow	Adjacent	Paper-trading tournaments	Live	No stakes, no fills, nothing to settle.
Pumpcade / Polymarket	Substitute	Prediction markets on events and stream moments	Large (Polymarket)	Betting on others, not competing yourself.
pump.fun bounties / livestreams	Channel	Pay-anyone-to-do-anything bounties, streams, creator fees	Record \$20M in creator earnings over seven days [S19]; \$7.9M over seven days in January 2026	Pay-to-do, not head-to-head. The streamer need is a later channel, not the launch wedge.
CEX trading competitions	Web2	Custodial, KYC, weeks-long, sponsored	Standard growth tactic	Not ten-minute social duels.
Melee Markets	Prospective venue / possible competitor	Permissionless prediction markets, 2 to 32 outcomes, bonding curve	\$3.5M seed led by Variant, DBA co-invested, angels incl. Anatoly Yakovenko and Meltem Demirs (Sept 2025) [S20]	Venue for spectator “who wins?” markets if it stays permissionless. If it ships live trading sessions it competes. Mitigation: Melee needs content and duels are content. Native spectator markets on Rug Royale remain a v3 decision gated on counsel. No contact yet.

Differentiation statement. Rug Royale is the only design that combines real spot fills, escrowed stakes, head-to-head settlement, and a coin-community rivalry with holder drip.

Nothing funded or shipped runs 1v1 duels with escrowed stakes on real spot fills of memecoins or StonkFun coins as of 2026-09-11.

PMF adjustment from the competitor check. Three changes. First, the landscape sorted names into partners and competitors: Melee and StonkFun are venues that need content; fomo and pump.fun are the fast-follow risk. Second, the fast-follow risk is a funded incumbent, so PMF proof must come from the first StonkFun cohort quickly rather than from a long build; the validation tests in Section 2 move earlier. Third, SolDuel's zero and PvPBattle's 30-second battles together mean the format must show skill, not just escrow, so the unranked on-ramp and the skill backtest are load-bearing, not optional.

4. Founder-Market Fit

Team. Two builders from the Turbin3 cohort. I lead product, data, frontend, and go-to-market. Vineet (RustRogue), a Rust and Solana engineer who built Conduit, leads the Anchor program. I have shipped full-stack Solana products for six months, including solenrich.com, and am deepening my Anchor skills in this cohort; I own the frontend, the data pipeline, the tests around the program, and the go-to-market, and I use AI tooling to move fast on all of it.

Why this problem. I got into crypto in 2014 after watching virtual gold get mined and traded in World of Warcraft: money was already a game. I have been a gamer since I was seven and a trader and builder onchain since 2021. I was an early participant in pump.fun, fomo, and StonkFun, and I watched entertainmentFi form from the ground up. The insight behind Rug Royale is that the game-theory mechanics of crypto apply to the interactions between traders themselves, not only to the tokens. Every trench chat already runs on call-outs and PnL flexes; nobody has given that game a referee and a pot.

Background.

- Crypto since 2014. Built and ran a BTC, LTC, and Dash mining business in 2017 to 2018 that earned over \$200K. Chief operating officer of a DAO in 2022. Ran a web3 consulting company from 2021 to 2024 helping clients build, launch, and improve onchain applications.
- Data analyst since 2015. Advanced analytics in Python (Springboard, 2021). Coding since 2019, full-stack dapps since 2021.
- 111words, a Farcaster miniapp that turns a daily writing habit into tradeable coins. Top prize at the Encode hackathon.
- SolEnrich (solenrich.com), an x402 agentic intelligence platform for agents trading on Solana. Finalist in the Bags hackathon; over 30 unique users and 700 transactions since launch.

How the background maps to this product's risks.

Risk or need in this LOI	What retires it
Skill at short windows is the top unvalidated assumption; the fix is a backtest	Ten years as a data analyst with Python. I run the backtest myself.
The entertainmentFi thesis has to be more than a slogan	111words already turned a social behavior into tradeable coins once.

Risk or need in this LOI	What retires it
Reward-mode drip, yield, and community incentives on StonkFun	Mining-era yield economics and a year running DAO operations.
Trading-bot behavior, ranked-mode integrity, Phase 2 agent entry	SolEnrich's wallet-graph and trading-agent data; agents enter the same duels via x402 with SolEnrich as the data layer.
The Anchor program with escrow, internal AMM, and settlement	Vineet leads the program; I have shipped full-stack dapps since 2021 and launched onchain products for consulting clients.

Network and distribution. Distribution is the hard part of this product, and these are the channels I can reach today: the StonkFun and Bags communities, where I am an early participant; X at @oxSardius with over 2,300 followers across crypto including people at major Solana protocols; the Turbin3 cohort as first duelists; and the Farcaster audience from 111words. The first StonkFun duel night and the first creator conversations go through these.

Focus. I build agents onchain and use agents to build onchain. Rug Royale's capstone is the human arena; agents join it in Phase 2. Both sides of my work point at the same product.

Code for 111words and SolEnrich is at github.com/oxSardius; SolEnrich is live at solenrich.com. The Encode and Bags hackathon listings have since been taken down by the organizers, so those results are stated here without links.

Phase 2: Use Cases & On-Chain Requirements

Capstone scope: devnet, internal deterministic market, modes unranked, freeroll, and friendly. Design choices: per-player coin with a shared quote asset; simulated independent liquidation at settlement; Token-2022 interface; account initialization split between `create_duel` and `join_duel`; `sponsor_prize`; permissionless `emergency_refund`; Duel account as the oracle; tunables in `Config`; optional settler tip.

StonkFun mapping. In the capstone, `quote_mint` is a program-minted devnet asset standing in for STONK, and `allowed_mints` holds demo coins standing in for graduated StonkFun coins, one of them with a 3% Token-2022 transfer fee to model reward mode. On mainnet, `quote_mint` is STONK, the allowlist is curated from the StonkFun public API, and valuation reads Raydium CPMM pool state instead of the internal pools.

Part A: Actors

Direct actors (signers who invoke instructions)

- **Creator** signs `create_duel` (picks coin A, tier, window, escrows entry).
- **Opponent** signs `join_duel` (picks coin B, escrows entry).
- **Duelist** (Creator or Opponent) signs `swap` during the window, on their own coin only.
- **Sponsor** signs `sponsor_prize` to fund the pot of an Open duel (freeroll).
- **Settler** is any signer who cranks `settle` after `end_ts`; may receive the optional tip.
- **Any signer** may call `cancel_duel` (Open past deadline), `emergency_refund` (Active past end plus grace), and `close_duel` (terminal). Funds always route to the parties on record, never to the caller.

- **Admin** signs `init_config` once. No other instruction requires the admin.

Beneficiaries (gain value without signing)

- **Winner** receives the pot inside the settler's transaction.
- **Treasury** receives rake inside `settle`.
- **Both players** receive refunds and rent when another party signs.
- **Sponsor** receives the sponsored amount back on cancel or refund.
- **Coin communities** receive swap volume and, on mainnet, reward-mode drip from duel activity.
- **Spectators** read vault balances and the Duel account for live PnL (client-side).

Administrators

- **Config admin:** a single key that runs `init_config` in the capstone. Upgrade path: multisig or governance PDA, Phase 6.
- **Treasury:** a pubkey in `Config`; recipient only.

Stakeholders (external interest, no transactions)

- **StonkFun** (launchpad; fees and STONK buyback benefit from duel volume), **coin creators**, **Raydium** (CPMM pools on mainnet), **prediction-market venues and indexers** (Melee, Helius) reading the Duel account, **Turbin3**, **SolEnrich** (integrity data, off-chain).

Part B: Use cases (one use case = one atomic state transition = one instruction handler)

Fields: Actor (signer) · Preconditions · Inputs · State transition · Accounts written · Transfers · Event · Typed errors.

UC-1 `init_config`

- **Actor:** Admin (signer, payer).
- **Preconditions:** `Config` does not exist.
- **Inputs:** `treasury`, `rake_bps`, `grace_secs`, `settler_tip_lamports` (0 allowed), `tiers: [u64; 3]` (bankroll per tier), `windows: [u32; 4]` (allowed window lengths), `pool_seed_ratio`, `quote_mint`, `allowed_mints: [Pubkey; 8]` (capstone allowlist).
- **State transition:** `none` → `Config`.
- **Accounts written:** `Config` (init).
- **Transfers:** rent.
- **Event:** none.
- **Errors:** `RakeTooHigh`, `InvalidWindowSet`.

UC-2 `create_duel`

- **Actor:** Creator (signer, payer).
- **Preconditions:** `Config` exists; `nonce` unused; `coin_a` ∈ `allowed_mints`; `window_secs` ∈ `Config.windows`; tier valid.

- **Inputs:** `nonce`, `tier`, `window_secs`, `coin_a`, `allowed_opponent: Option<Pubkey>`, `join_deadline`.
- **State transition:** `none` → `Duel { status: Open, creator, coin_a, quote_mint, bankroll(tier), window_secs, join_deadline, entry_lamports }`.
- **Accounts written:** `Duel` (init), `Escrow` (init), creator vault authority and its quote and coin-A token accounts (init, Token-2022 interface), `Pool[coin_a]` and its token accounts (init, seeded from `pool_seed_ratio`).
- **Transfers:** `entry_lamports` `Creator` → `Escrow` (0 in unranked and freeroll); pool seed minted by the `MintAuthority` PDA.
- **Event:** `DuelCreated`.
- **Errors:** `InvalidTier`, `InvalidWindow`, `MintNotAllowed`, `DeadlineTooSoon`, `DeadlineTooFar`.

UC-3 `sponsor_prize`

- **Actor:** `Sponsor` (signer).
- **Preconditions:** `status == Open`; `Duel.sponsor` is unset or equals the signer.
- **Inputs:** `amount`.
- **State transition:** `Duel.sponsored_lamports += amount`, `Duel.sponsor = signer`. Status unchanged.
- **Accounts written:** `Duel`, `Escrow`.
- **Transfers:** `amount` `Sponsor` → `Escrow`.
- **Event:** `PrizeSponsored`.
- **Errors:** `DuelNotOpen`, `SponsorMismatch`, `ZeroAmount`.

UC-4 `join_duel`

- **Actor:** `Opponent` (signer, payer).
- **Preconditions:** `status == Open`; `now < join_deadline`; `signer != creator`; `allowed_opponent` matches if set; `coin_b` ∈ `allowed_mints`.
- **Inputs:** `coin_b` (may equal `coin_a`).
- **State transition:** `Open` → `Active`; sets `opponent`, `coin_b`, `start_ts = now + 60`, `end_ts = start_ts + window_secs`.
- **Accounts written:** `Duel`; opponent vault authority and its quote and coin-B token accounts (init); `Pool[coin_b]` and token accounts (init only if `coin_b != coin_a`); both quote vaults credited with the bankroll.
- **Transfers:** `entry_lamports` `Opponent` → `Escrow`; bankroll minted by `MintAuthority` into both quote vaults; pool seed for coin B if new.
- **Event:** `DuelJoined`.
- **Errors:** `DuelNotOpen`, `JoinDeadlinePassed`, `CannotJoinOwnDuel`, `OpponentNotAllowed`, `MintNotAllowed`.

UC-5 `swap`

- **Actor:** `Duelist` (signer; equals `creator` or `opponent`).
- **Preconditions:** `status == Active`; `start_ts <= now < end_ts`; the pool is the signer's own coin.
- **Inputs:** `side: Buy | Sell`, `amount_in`, `min_out`.

- **State transition:** `Pool.{quote_reserve, token_reserve}` updated by constant-product math with fee; vault balances updated. Status unchanged.
- **Accounts written:** `Pool`, `player quote vault`, `player coin vault`, `pool quote` and `coin accounts`.
- **Transfers:** PDA-signed `transfer_checked` (Token-2022 interface; transfer-fee mints supported) between vault and pool.
- **Event:** `SwapExecuted { duel, player, mint, side, in_amt, out_amt }`.
- **Errors:** `DuelNotActive`, `WindowNotStarted`, `WindowEnded`, `NotAParticipant`, `WrongPool`, `InsufficientBankroll`, `SlippageExceeded`, `MathOverflow`.

UC-6 `settle`

- **Actor:** Settler (any signer).
- **Preconditions:** `status == Active`; `now >= end_ts`.
- **Inputs:** none.
- **State transition:** `Active` → `Settled`. For each player: `final = quote_vault_balance + simulated_sell(coin_vault_balance, own pool reserves)`; no pool mutation. `pnl = final / bankroll`. Records `winner`, `final_a`, `final_b`. Tie if equal.
- **Accounts written:** `Duel`, `Escrow`, `Treasury`, `winner wallet`, `settler wallet (tip)`, both `players (tie)`, `sponsor (tie)`.
- **Transfers:** `pot = 2 × entry + sponsored`; `rake = pot × rake_bps` (0 when entry is 0); `tip = min(settler_tip_lamports, pot - rake)`; winner receives `pot - rake - tip`. Tie: entries back to each player, sponsored split equally, no rake, tip still paid if configured.
- **Event:** `DuelSettled { duel, winner, final_a, final_b, prize }` (notification only; the `Duel` account is the record).
- **Errors:** `DuelNotActive`, `WindowNotEnded`, `MathOverflow`.

UC-7 `cancel_duel`

- **Actor:** Any signer.
- **Preconditions:** `status == Open`; `now > join_deadline`.
- **State transition:** `Open` → `Cancelled`.
- **Accounts written:** `Duel`, `Escrow`.
- **Transfers:** `entry` → `Creator`; `sponsored` → `Sponsor`.
- **Event:** `DuelRefunded { reason: NoOpponent }`.
- **Errors:** `DuelNotOpen`, `DeadlineNotReached`.

UC-8 `emergency_refund`

- **Actor:** Any signer.
- **Preconditions:** `status == Active`; `now > end_ts + grace_secs`.
- **State transition:** `Active` → `Refunded`.
- **Accounts written:** `Duel`, `Escrow`.
- **Transfers:** `entry` → each player; `sponsored` → `Sponsor`.
- **Event:** `DuelRefunded { reason: SettlementNotCranked }`.
- **Errors:** `DuelNotActive`, `GraceNotElapsed`.

UC-9 close_duel

- **Actor:** Any signer.
- **Preconditions:** `status ∈ {Settled, Cancelled, Refunded}`.
- **State transition:** vaults, pools, and Escrow closed. The Duel account stays open as the permanent record (oracle). Sets `Duel.closed = true`.
- **Accounts written:** vaults, pools, Escrow (closed); Duel.
- **Transfers:** rent → the party that paid for each account (creator for shared accounts and their vaults; opponent for theirs).
- **Event:** none.
- **Errors:** `DuelStillLive`, `AlreadyClosed`.

Adversarial analysis and granularity check (final set)

Rule	Check	Result
1. Atomicity	Nine handlers, nine transitions: <code>create</code> (none→Open), <code>sponsor</code> (pot funded), <code>join</code> (Open→Active), <code>swap</code> (pool and vault state), <code>settle</code> (Active→Settled), <code>cancel</code> (Open→Cancelled), <code>emergency_refund</code> (Active→Refunded), <code>close</code> (cleanup), <code>init_config</code> (none→Config). The former <code>refund</code> was split; account initialization moved to <code>create</code> / <code>join</code> so no handler bundles two lifecycle steps.	Pass
2. State ownership	Duel state, results, pools, vaults, escrow, sponsor, and config are program-owned accounts. Points, lobby listing, PnL curves, result cards, allowlist curation, and collusion signals are declared client-side or indexer. The Duel account is the oracle; the event is a notification.	Pass
3. Real signers	Every instruction names its signer. Admin appears only in <code>init_config</code> . Refund paths, settlement, and close are permissionless with funds routed to parties on record. Play-money bankroll is minted by a program PDA, not a script key; demo coin mints have authority burned or PDA-held. Tunables live in <code>Config</code> , not in a backend.	Pass
4. On-chain vs client-side	Protocol: escrow, bankroll issuance, swaps, valuation, payout, refunds, config. Client: timers, PnL bars (RPC polling of vaults), result cards and OG images, lobby (<code>getProgramAccounts</code> on devnet; indexer on mainnet), points and unranked leaderboard, StonkFun allowlist curation from the public API, host links, SolEnrich integrity signals.	Pass

Design notes: reward-mode drip cannot be reproduced on devnet (Token-2022 withheld fees go to a fee authority, not holders), so the capstone models the tax only; on mainnet, any drip received by a vault before `settle` counts in `final`. Settler tip stays optional (`settler_tip_lamports = 0` disables it).

Consolidated on-chain requirements matrix (capstone)

Accounts and PDAs

Account	Seeds	Owner	Purpose
Config	<code>["config"]</code>	program	admin, treasury, <code>rake_bps</code> , <code>grace_secs</code> , <code>settler_tip</code> , tiers, windows, <code>pool_seed_ratio</code> , <code>quote_mint</code> , <code>allowed_mints</code>
MintAuthority	<code>["mint_authority"]</code>	program	mint authority of the devnet quote mint (bankroll and pool seed)
Duel	<code>["duel", creator, nonce_le]</code>	program	state machine, players, <code>coin_a</code> , <code>coin_b</code> , <code>quote_mint</code> , window, entries, sponsor, results; permanent record

Account	Seeds	Owner	Purpose
Escrow	["escrow", duel]	system (PDA-signed)	entries and sponsored lamports
Vault authority	["vault", duel, player]	program	signs transfers for one player
Player quote vault, player coin vault	ATA(vault authority, mint), Token-2022 interface	Token program	bankroll and position
Pool (1 or 2 per duel)	["pool", duel, mint]	program	constant-product reserves for one coin
Pool quote and coin accounts	ATA(pool, mint), Token-2022 interface	Token program	reserves

Instruction handlers (9): `init_config`, `create_duel`, `sponsor_prize`, `join_duel`, `swap`, `settle`, `cancel_duel`, `emergency_refund`, `close_duel`.

CPI dependencies: System Program (lamport transfers, account creation); Token-2022 and SPL Token via `token_interface` (`mint_to`, `transfer_checked`, `close_account`); Associated Token Program. No external DEX CPI in the capstone; mainnet replaces internal pools with Raydium CPMM reads or Jupiter fills (Phase 6).

Custom constraints

- `rake_bps <= 1000`; rake is 0 when `entry_lamports == 0`.
- `window_secs ∈ Config.windows`; tier maps to `Config.tiers[tier]`.
- `coin_a, coin_b ∈ Config.allowed_mints`; neither equals `quote_mint`.
- `join_deadline ∈ [now + 60, now + 86400]`.
- `swap` only on the signer's own coin; `amount_in <= vault balance`; constant-product with fee so `k` never decreases; `min_out` enforced.
- Settlement valuation is a pure function of vault balances and pool reserves (simulated sell); no order dependence.
- One sponsor per duel; sponsored funds return to the sponsor on cancel or refund; split equally on tie.
- Every transfer PDA-signed with explicit seeds; checked math; no `unwrap()`; Token-2022 `transfer_checked` with fee-aware amounts.
- First valid `join_duel` wins the seat (status gate).
- `settle`, `cancel_duel`, `emergency_refund`, `close_duel` are permissionless; the Duel account is never closed.

Explicitly off-chain

- Points, unranked leaderboard, lobby listing, live PnL bars, timers, result cards, share links.
 - StonkFun allowlist curation (graduated, standard-mode preferred, reserve floor) from the public API; written into `Config.allowed_mints` by the admin on devnet, Merkle root or allowlist account on mainnet (Phase 6).
 - Streamer host links and pot share (Phase 6; on-chain `host` field when built).
 - Collusion signals from SolEnrich (advisory; gates ranked matchmaking off-chain).
-

Sources (Part 1 citations)

#	Source	Used for	Accessed
S1	Crypto Briefing, “Fomo and Pump.fun battle for trader flow in Solana’s memecoin arms race”, https://cryptobriefing.com/fomo-pumpfun-solana-memecoin-competition/	fomo weekly revenue > \$2M; overtook Axiom in daily fees 2026-08-06; pump.fun social trading feature 2026-08-07; pump.fun DAU ATH	2026-09-09
S2	blocmates, “fomo: A Mobile-First Social Trading Experience”, https://www.blocmates.com/articles/fomo-a-mobile-first-social-trading-experience ; Surf AI, “How fomo became Solana’s social trading hotspot through giveaways and leaderboards”, https://asksurf.ai/pulse/en/fomo-virality-solana-social-breakout ; FomoVille, https://fomoville.town/	fomo sponsored PnL competitions (e.g., \$10K of \$SKATE), leaderboards; FomoVille 2.5D prize city with weekly prize-pool payouts	2026-09-11
S3	The Block, “STONK surges 250% to \$140 million market cap as stock-paired Solana launchpad StonkFun pulls volume to Raydium and Jupiter” (2026-09-06), https://www.theblock.co/news/defi/2026-09-06-stonk-surges-250-to-140-million-market-cap-as-stock-paired-solana-launchpad-stonkfun-pulls-volume-to-raydium-and-jupiter-413621	STONK ~\$140M market cap, ~\$135M daily volume, Raydium LaunchLab integration	2026-09-11
S4	StonkFun, Rewards, https://www.stonkfun.xyz/rewards	Reward-mode transfer tax (1% / 3%) paid to holders in the quote asset	2026-09-11
S5	stonk.fyi (unofficial STONK dashboard), https://stonk.fyi/	~55.2K holders, ~1.3K tokens, ~\$7.62M fees/week, 14.48% burned	2026-09-11
S6	StonkFun, Developer API, https://www.stonkfun.xyz/developers	Free public API, no key, 300 req/min; launch tooling (dev buys up to 50%, airdrops, reward mode); no competition/leaderboard/PnL features	2026-09-11
S7	The Stonk Board, https://thestonkboard.com/	Only third-party StonkFun tool found; reward-coin dashboard, no games	2026-09-11
S8	SolDuel, https://solduel.xyz/	Model; 0 volume / 0 matches / 0 traders	2026-08-31
S9	TradeFighters, https://tradefighters.fun/	Product confirmed by manual check; general Solana trading duels with leaderboard	2026-09-11
S10	PerpClash, https://perpclash.com/	Timed 1v1 perps duels, no deposit, season prize vault (search snippet); site 404 on fetch and not live per manual check	2026-09-11
S11	PvPBattle, https://www.pvpbattle.xyz/	Virtual positions vs Binance mark, custodial, 30s–15min, up to 25x, example pots 0.5–4 SOL	2026-09-09

#	Source	Used for	Accessed
S12	Solana Compass, “FOMO Social Trading App Ranks Third in Solana 7-Day Protocol Revenue”, https://solanacompass.com/news/fomo-social-trading-app-holds-third-in-solana-protocol-revenue-for-a-full-week	fomo third in 7-day protocol revenue	2026-09-11
S13	insights4.vc, “fomo Crypto Trading App: \$75M Series B, Product and Strategy”, https://insights4.vc/blog/fomo-behind-the-75-series-b/	fomo \$75M Series B, \$550M valuation, Index Ventures lead, USV; 625,000 registered traders, ~3,500/day	2026-09-11
S14	Duel Duck, https://duelduck.com/	Yes/no prediction duels collateralized in any SPL token	2026-09-11
S15	a16z crypto, “Investing in Kairos”, https://a16zcrypto.com/posts/article/investing-in-kairos/	Kairos seed; tournament details from search	2026-09-11
S16	Big Dance, https://www.bigdance.fun/	Tournament prediction market; 50% referral share	2026-09-11
S17	Solana Media, “Solana Ecosystem Roundup: March 2026”, https://solana.com/news/solana-ecosystem-roundup-march-2026	Duel Duck SPL-token predictions; Kairos TradeX, Big Dance, Scorely launches; PlayBabylon agent activity	2026-09-11
S18	Solana Compass, Colosseum Frontier Hackathon 2026 winners, https://solanacompass.com/news/colosseum-announces-26-winners-of-the-solana-frontier-hackathon-the-largest-crypto-hackathon-ever	No trading-duel project among winners	2026-09-09
S19	CryptoSlate, “Pump.fun’s PUMP revitalized by streaming success and record payouts”, https://cryptoslate.com/pumpfun-memecoin-streams-explode-as-creators-pocket-record-earnings-in-last-week/	Record \$20M creator earnings in seven days; \$7.9M/7d in Jan 2026 (Yahoo Finance / CoinMarketCap coverage of Project Ascend fee model)	2026-09-11
S20	CoinDesk, “Melee Raises \$3.5M to Launch ‘Viral Prediction Markets’ Without Gatekeepers” (2025-09-24), https://www.coindesk.com/business/2025/09/24/melee-raises-usd3-5m-to-launch-viral-prediction-markets-without-gatekeepers	Melee seed: Variant lead, DBA, angels incl. Anatoly Yakovenko, Meltem Demirors	2026-09-11

Part 2: Process Appendix (Red Team Log)

This appendix is the proof of work: the original drafts, the AI critique, the founder's rebuttals, and the decisions, recorded verbatim. Section A reproduces the original Draft v1 that the opening attack tables target. Section B is the log itself, including the dialogue rounds.

A. Original Draft v1 (pre-red-team)

Rug Royale: Turbin3 LOI, Draft v1 (pre-red-team)

Status: ORIGINAL DRAFT. This is the “before” document for the Process Appendix. It is compiled from the research and design work in this folder (shortlist 2026-08-25, deep dives v1-v3, PRD, LOI source). The red team runs against this text in docs/turbin3-red-team-log.md; the refined version becomes Part 1 of the submission.

Items marked [FOUNDER INPUT] need facts only the founder has. Do not submit with placeholders.

Project overview

Rug Royale is a 1v1 PvP trading-duel protocol on Solana. Two traders lock an entry into a trustless escrow, each trades an equal, fixed bankroll on an agreed token list for a fixed window, and the program settles the winner from verifiable onchain fills. PnL flexing in memecoin trading today runs on screenshots that nobody can verify, settle, or monetize; Rug Royale makes the flex settleable. The capstone ships the escrow, swap, and settlement program on devnet with a thin duel-room frontend. Post-capstone, the same settlement state becomes a public resolution oracle for spectator markets and generalizes to N-player battle-royale tournaments.

Phase 1: Project Definition & Market Analysis

Part A.1: Core Value Proposition & Product-Market Fit

Value proposition (one paragraph). Rug Royale turns the most common social behavior in memecoin trading, proving you are the better trader, into a game with real stakes and a trustless referee. Two players lock an entry (E) into escrow and trade an identical bankroll (B) for 5, 15, or 60 minutes; at the buzzer, positions are force-sold to the quote asset and the higher PnL% takes the prize (2E minus rake). Separating E from B bounds each player's loss, keeps PnL% comparable across players, and creates two revenue streams: a rake on the prize and integrator fees on every swap. Because escrow, fills, and settlement all happen onchain, results cannot be faked, disputes cannot arise, and no bookmaker or broker is needed. In Web2 this product cannot legally exist without a brokerage license and a gaming license; on Solana it is one Anchor

program plus audited DEX infrastructure. The tagline: “Stop flexing screenshots. Settle it onchain.”

Why now.

- Social trading on pump.fun streams is a measurable behavior: streamer tokens did about \$1.6B volume over 30 days, and pump.fun pays roughly \$20M per week in creator fees. Streamers need new interactive segments.
- **fomo** runs sponsored many-player PnL competitions and reports 110M social interactions. Demand for competitive trading is proven; a head-to-head, stake-your-own format is not served.
- Melee Markets (permissionless prediction markets, \$3.5M seed) now exists as an external venue for spectator “who wins?” markets, so the duel can be the content without the protocol running a book.

PMF hypothesis. The first retained cohort is on-stream degens who already argue about PnL in pump.fun and X chats. The wedge is streamer-hosted duels: a streamer hosts viewer-vs-viewer duels as a segment and earns 1% of every pot. The retention hook is the shareable result card with two PnL curves and transaction links.

Coverage check, value drivers listed vs. possibly missing.

Driver	Covered?	Note
Verifiable outcome (trustless referee)	Yes	Core
Bounded loss (E separate from B)	Yes	Core
Shareable content (result card, live PnL bars)	Yes	Retention
Streamer revenue share	Yes	Distribution
Spectator market hook (DuelSettled as oracle)	Yes	Post-capstone
Skill measurement / rating over time	Partial	“Ladder” ranked seasons are Phase 6 only
Learning value (replay a better trader’s fills)	Not covered	Possible driver: post-settle replay of both players’ fills
Agent entry (bot vs bot)	Mentioned	Phase 2, not a v1 driver
Sponsor / brand value (branded freerolls)	Not covered	Possible driver for freeroll funding

Part A.2: Target Markets & User Profiles

Target market segments (2–5).

1. Stonkfun eco participants who want to trade between their community coins within the eco, or dface off against traders of other ecos. Stonk side has a beneficial piece where the transfer fees let a rising tide raise all boats.
2. **Solana memecoin retail traders** who trade daily on pump.fun, Axiom, Photon, or BullX and post PnL. Largest segment; primary players.

3. **pump.fun and X livestreamers** who monetize through creator fees and need interactive formats. Distribution channel and secondary revenue recipients.
4. **Trading-content spectators** who watch trench streams and bet on outcomes elsewhere (Polymarket, Pumpcade). Audience and, via Melee, the “handle.”
5. **Trading-agent builders** (Solana Agent Kit, elizaOS, Colosseum teams) who want a fair public arena for bots. Phase 2.
6. **Sponsors and token teams** who fund competitions as marketing (as fomo’s sponsors do today). Freeroll prize funders.

User profiles (top 3).

Profile	Who	Job to be done	Today’s workaround	Why Rug Royale
The on-stream degen	Trades memecoins daily, active in stream chats, posts PnL screenshots, bankroll \$100–\$5k	Prove I am the better trader and win money doing it	Screenshots, call-outs in chat, unverifiable bets between friends	Onchain proof, bounded stake, instant settlement, shareable card
The streamer / host	Top-100 pump.fun or X trench streamer, earns creator fees, needs segments that hold viewers	Keep chat engaged, add a revenue line	Bounties, giveaways, sponsored comps	Host link routes 1% of every pot to the host; live PnL overlay for OBS
The spectator	Watches trench streams nightly, follows 5–10 traders, bets on Polymarket occasionally	Have a rooting interest in a live event	Watch and comment; no stake	Live PnL bars; later, a “who wins?” market on Melee resolved by the duel’s own settlement

Coverage check against ecosystem demand.

- Evidence for segment 1: fomo’s competition participation and 625k users; daily PnL-screenshot volume on X.
- Evidence for segment 2: \$20M/week pump.fun creator payouts; streamers already run bounties and callouts (the reverse-engineered pumpfun-apis spec shows livestream, bounties v2, callouts, and fee-sharing endpoints).
- Evidence for segment 3: Pumpcade and Polymarket markets on stream moments; Melee’s raise.
- Weak evidence: segment 4 (agent builders) and segment 5 (sponsors) are assumptions from adjacent products, not from direct conversations.
- Not yet tested: whether any segment will pay entry rather than only play freerolls. The 7-day validation sprint (streamer DMs, mock result card, manual duels) was designed for this and has not been run.

Part A.3: Competitor Landscape

Primary research sources used so far: product sites (solduel.xyz, pvpbattle.xyz, melee.markets, fomoville.town, pumpportal.fun), GitHub (BankkRoll/pumpfun-apis, TeamRaccoons/referral), X and blocmates/asksurf coverage of fomo, Crunchbase/ChainCatcher for the Melee seed. Verified 2026-08-31. Not yet run: DefiLlama raises tracker query, Colosseum Copilot search for hackathon precedents.

Product	Type	Model	Traction signal	Gap vs Rug Royale
SolDuel	Direct	Pick-a-token, best % change wins; onchain escrow; mainnet, audited	0 volume, 0 matches, 0 traders	No trading, no skill expression. Proves escrow alone attracts nobody.
PvPBattle	Direct	Paper trades vs Binance mark price; custodial SOL wagers	Live, unknown volume	Simulated fills = price wagering; nothing verifiable onchain; weaker legal posture
fomo competitions / FomoVille	Substitute	Sponsored many-player PnL competitions, leaderboards, prize city	625k users, 110M social interactions	No stakes, no head-to-head. Could add 1v1; speed matters.
DUELPVP / LootScape	Adjacent	Dice, coinflip, VRF for SOL	Live	Chance, not skill; no content angle
TradeFlow	Adjacent	Paper-trading tournaments	Live	No stakes, no fills, nothing to settle
Pumpcade / Polymarket	Substitute	Prediction markets on events and stream moments	Large (Polymarket)	Betting on others, not competing yourself
pump.fun bounties / livestreams	Adjacent	Pay-anyone-to-do-anything bounties, streams	\$20M/week creator payouts	Pay-to-do, not head-to-head; the streamer need is the wedge
CEX trading competitions	Web2	Custodial, KYC, weeks-long, sponsored	Standard growth tactic	Not 15-minute social duels
Melee Markets	Partner	Permissionless prediction markets, 2–32 outcomes, bonding curve	\$3.5M seed (Variant; Toly angel)	Venue for spectator markets, not a competitor

Initial differentiation statement. Rug Royale is the only design that combines real DEX fills, real stakes, and head-to-head trustless settlement. SolDuel’s zero shows that escrow alone is not a product; distribution through streamers is the plan for the missing piece.

Part A.4: Founder-Market Fit

[FOUNDER INPUT, rewrite in first person with real facts before red-teaming]

- **Background:** Active Solana memecoin ecosystem participant. [years trading, communities, notable activity]
- **Technical skills:** Builder of SolEnrich (solenrich.com), a live x402-paid onchain-data API with wallet-graph, copy-trade analytics, and risk scoring. [languages, Anchor/Rust experience level, frontend stack, prior shipped projects]
- **Why this problem:** [personal story: the screenshot argument, the stream moment, the friend bet that could not be settled]
- **Network:** [streamers, traders, Turbin3 cohort, Superteam, Colosseum contacts, agent-builder communities]
- **Unfair advantage claimed:** SolEnrich’s wallet-graph and risk scoring become Rug Royale’s integrity layer (collusion-cluster detection, duelist credibility scores, token allowlist), which no competitor in the table has.
- **Team:** [solo or team; roles]. Assumed solo builder for the capstone.

Phase 2: First Use Cases & On-Chain Requirements (capstone scope)

Part A: Actors

Direct actors (signers who invoke instructions)

- **Creator** signs `create_duel`; escrows the first entry; may sign `cancel_duel` after the join deadline.
- **Opponent** signs `join_duel`; escrows the second entry.
- **Duelist** (Creator or Opponent) signs `swap` during the window.
- **Settler (crank)** is any wallet that signs `settle` after `end`. Permissionless.
- **Closer** is any wallet that signs `close_duel` after a terminal state. Rent returns to the players, not the closer.
- **Admin** signs `init_config`; signs `emergency_refund` on stuck Active duels after the grace period.

Beneficiaries (gain value without signing)

- **Winner** receives the prize inside the settler's transaction.
- **Treasury** receives rake inside `settle`.
- **Both players** receive rent back inside `close_duel`, and refunds in `cancel_duel` / `emergency_refund` when another party signs.
- **Spectators** read vault balances and events for live PnL (client-side).
- **Streamer / host** (post-capstone) receives a pot share.

Administrators

- **Config admin** is a single admin key on the `Config` PDA (capstone). A governance PDA or multisig is a Phase 6 upgrade.
- **Treasury** is a pubkey stored in `Config`. Recipient only, no authority.

Stakeholders (external interest, no transactions)

- **Prediction-market venues and indexers** (Melee, Helius) consume `DuelSettled` as a resolution oracle.
- **Streamers** are content and distribution partners.
- **Sponsors** fund freeroll prizes.
- **Turbin3** evaluates the capstone.
- **SolEnrich** provides integrity data (off-chain).

Part B: Use Cases (one use case = one atomic state transition = one instruction handler)

Template fields: Actor (signer) · Preconditions · Inputs · State transition · Accounts written · Transfers · Event · Typed errors.

UC-1 `init_config`

- **Actor:** Admin (signer, payer).

- **Preconditions:** Config PDA does not exist.
- **Inputs:** rake_bps: u16, treasury: Pubkey, grace_secs: u32.
- **State transition:** none $\rightarrow$ Config { admin, rake_bps, treasury, grace_secs }.
- **Accounts written:** Config (init).
- **Transfers:** rent only.
- **Event:** none.
- **Errors:** RakeTooHigh (rake_bps > 1000).

UC-2 create_duel

- **Actor:** Creator (signer, payer).
- **Preconditions:** Config exists; nonce unused for this creator.
- **Inputs:** nonce: u64, tier: u8, window_secs: u32, token_list: [Pubkey; 3], allowed_opponent: Option<Pubkey>, join_deadline: i64.
- **State transition:** none $\rightarrow$ Duel { status: Open, creator, entry, bankroll(tier), window, token_list, join_deadline }.
- **Accounts written:** Duel (init), Escrow (init, SystemAccount PDA).
- **Transfers:** entry_lamports Creator $\rightarrow$ Escrow (0 in freeroll).
- **Event:** DuelCreated.
- **Errors:** InvalidTier, InvalidWindow, DuplicateMint, DeadlineTooSoon, DeadlineTooFar.

UC-3 join_duel

- **Actor:** Opponent (signer, payer).
- **Preconditions:** Duel.status == Open; now < join_deadline; Opponent != Creator; allowed_opponent matches if set.
- **Inputs:** none.
- **State transition:** Open $\rightarrow$ Active; sets opponent, start_ts = now + 60, end_ts = start_ts + window.
- **Accounts written:** Duel; vault authority PDAs and their token accounts for both players (init); Pool PDAs and pool token accounts for each of the 3 mints (init, seeded from config constants); bankroll minted or transferred into both quote vaults.
- **Transfers:** entry_lamports Opponent $\rightarrow$ Escrow; bankroll B into each player's quote vault; seed reserves into each pool.
- **Event:** DuelJoined.
- **Errors:** DuelNotOpen, JoinDeadlinePassed, CannotJoinOwnDuel, OpponentNotAllowed.

UC-4 swap

- **Actor:** Duelist (signer; must equal creator or opponent).
- **Preconditions:** status == Active; start_ts <= now < end_ts; mint $\in$ token_list.
- **Inputs:** mint: Pubkey, side: Buy|Sell, amount_in: u64, min_out: u64.
- **State transition:** Pool.{quote_reserve, token_reserve} updated by constant-product math; player vault balances updated. Duel.status unchanged.
- **Accounts written:** Pool, player quote vault, player token vault, pool quote account, pool token account.
- **Transfers:** PDA-signed SPL transfers vault $\leftrightarrow$ pool.
- **Event:** SwapExecuted { duel, player, mint, side, in_amt, out_amt }.

- **Errors:** `DuelNotActive`, `WindowNotStarted`, `WindowEnded`, `NotAParticipant`, `TokenNotInList`, `InsufficientBankroll`, `SlippageExceeded`, `MathOverflow`.

UC-5 `settle`

- **Actor:** Settler (any signer).
- **Preconditions:** `status == Active`; `now >= end_ts`.
- **Inputs:** none.
- **State transition:** `Active` → `Settled`; force-sells each player's token balances into the pools; compares final quote balances; records `winner`, `final_a`, `final_b`.
- **Accounts written:** `Duel`, both players' vaults, all `Pools`, `Escrow`, `Treasury`, winner's wallet.
- **Transfers:** prize ($2E - \text{rake}$) `Escrow` → winner; rake `Escrow` → `Treasury`; on tie, E back to each player, no rake.
- **Event:** `DuelSettled` { `duel`, `winner`, `final_a`, `final_b`, `prize` }.
- **Errors:** `DuelNotActive`, `WindowNotEnded`, `AlreadySettled`.

UC-6 `cancel_duel`

- **Actor:** Creator (signer), or any signer with the refund always routed to Creator.
- **Preconditions:** `status == Open`; `now > join_deadline`.
- **Inputs:** none.
- **State transition:** `Open` → `Cancelled`.
- **Accounts written:** `Duel`, `Escrow`.
- **Transfers:** `entry_lamports` `Escrow` → Creator.
- **Event:** `DuelRefunded` { `reason: NoOpponent` }.
- **Errors:** `DuelNotOpen`, `DeadlineNotReached`.

UC-7 `emergency_refund`

- **Actor:** Admin (signer).
- **Preconditions:** `status == Active`; `now > end_ts + grace_secs`; settlement has not completed.
- **Inputs:** none.
- **State transition:** `Active` → `Refunded`.
- **Accounts written:** `Duel`, `Escrow`.
- **Transfers:** E `Escrow` → each player.
- **Event:** `DuelRefunded` { `reason: SettlementFailed` }.
- **Errors:** `DuelNotActive`, `GraceNotElapsed`, `Unauthorized`.

UC-8 `close_duel`

- **Actor:** Closer (any signer).
- **Preconditions:** `status ∈ {Settled, Cancelled, Refunded}`.
- **Inputs:** none.
- **State transition:** all duel-scoped accounts closed.
- **Accounts written:** `Duel`, `Escrow`, vaults, pools (all closed).
- **Transfers:** rent lamports → Creator and Opponent in proportion to what each paid.

- **Event:** none.
- **Errors:** `DuelStillLive`.

Note: the PRD lists 7 instructions with `refund` covering two paths. This draft splits it into UC-6 and UC-7 to satisfy the one-transition rule. The granularity check in the red team log confirms or rejects this split.

On-chain requirements: consolidated matrix (capstone)

Accounts and PDAs

Account	Seeds	Owner	Purpose
Config	<code>["config"]</code>	program	admin, rake_bps, treasury, grace_secs, pool seed constants
Duel	<code>["duel", creator, nonce_le_bytes]</code>	program	state machine, participants, window, token list, results
Escrow	<code>["escrow", duel]</code>	system (PDA-signed)	holds both entries in lamports
Vault authority	<code>["vault", duel, player]</code>	program (authority only)	signs SPL transfers for one player
Player quote vault	<code>ATA(vault authority, quote_mint)</code>	SPL Token	bankroll
Player token vaults ×3	<code>ATA(vault authority, mint_i)</code>	SPL Token	positions
Pool ×3	<code>["pool", duel, mint]</code>	program	constant-product reserves
Pool quote / token accounts ×3	<code>ATA(pool, quote_mint) / ATA(pool, mint)</code>	SPL Token	reserves

Instruction handlers (8): `init_config`, `create_duel`, `join_duel`, `swap`, `settle`, `cancel_duel`, `emergency_refund`, `close_duel`.

CPI dependencies: System Program (lamport transfers, account creation), SPL Token Program (transfer, `close_account`), Associated Token Program (ATA creation). No external DEX CPI in the capstone; Jupiter Ultra replaces the internal pools on mainnet (Phase 6).

Custom constraints

- `rake_bps <= 1000`; rake is 0 when `entry_lamports == 0` (freeroll).
- `window_secs ∈ {300, 900, 3600}`; tier maps to a fixed bankroll constant.
- `token_list` has 3 distinct mints, none equal to the quote mint.
- `join_deadline` between `now + 60` and `now + 86400`.
- `swap` rejects `amount_in` larger than the vault balance; constant-product with a fee so `k` never decreases; `min_out` enforced.
- Every transfer is PDA-signed with explicit seeds; checked math only; no `unwrap()`.
- First valid `join_duel` wins the seat (status gate provides atomicity).

- `settle` is permissionless and deterministic; tie refunds both without rake.

Off-chain / client-side (explicitly not protocol logic)

- Live PnL bars, timers, elimination-line UI: RPC polling of vault balances.
- Result cards, OG images, share links.
- Lobby listing of open duels: RPC `getProgramAccounts` or an indexer.
- Token allowlist curation and liquidity floors (mainnet).
- Streamer host links and pot-share routing (Phase 6; may move on-chain later).
- Collusion detection via SolEnrich wallet graph (advisory; may gate ranked matchmaking off-chain).

B. Red Team Log

Rug Royale: Red Team Log (Process Appendix, Part 2)

Target document: `docs/turbin3-loi-draft-v1.md` (Draft v1, 2026-09-09). Method: adversarial AI critique of each Phase 1 section, the FMF, and the Phase 2 use cases against the four granularity rules. Each attack has a proposed validity rating and a proposed decision. **Decisions marked PROPOSED are the AI’s recommendation; the founder confirms, overrides, or rejects each one and records the reason in the “Founder decision” column before the refined LOI is written.**

Ratings: **Valid** (accept, change the draft) · **Partially valid** (accept with a narrower change) · **Invalid** (reject, record why).

B.1: Red team: Value Proposition & PMF

#	Attack	Proposed validity	Proposed decision / change	Founder decision
VP-1	Verifiability is not a felt pain. Nobody loses sleep over an unverifiable screenshot. The felt motives are winning money and status. SolDuel had trustless escrow and verifiable results and got zero users. The draft leads with the referee, not the reason to play.	Valid	Accept. Lead the value prop with the game and the stakes (“prove you are better, get paid”). Move verifiability to the “why onchain” paragraph as the enabler, not the hook.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
VP-2	Skill may not exist at 5–15 minute horizons. If PnL% over a 5-minute window on three memecoins is close to a coin flip, the better trader does not win reliably, the pride motive fails, and the skill-contest legal defense collapses. The backtest that would settle this (validation sprint day 4) was never run.	Valid	Accept. Run the backtest before submission if possible; otherwise state it as the top unvalidated assumption and make the 15- and 60-minute windows the default, not 5.	
VP-3	The format removes most of memecoin skill. Same tokens, same bankroll, same window strips token selection and multi-hour timing, which is where real edge lives. What remains is scalping three pre-chosen mints.	Partially valid	Accept in part. Keep equal bankroll (needed for comparable PnL%). Let the creator choose the token list from a curated allowlist so token selection is part of the game. Consider a “pick 3 of 10” draft step post-capstone.	
VP-4	The capstone does not demonstrate the value prop. On devnet the game trades throwaway mints against a program-owned AMM with program-minted bankroll. That is a simulated market with escrow, structurally the same thing the draft criticizes PvPBattle for. “Real fills on real memecoins” is not shown on demo day.	Valid	Accept. State plainly in the LOI: the capstone proves the escrow, settlement, and state machine; real-market fills are the mainnet path (Jupiter). Frame the internal AMM as a deterministic test harness, not the product.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
VP-5	Shared pools create a settlement-order exploit. If settle force-sells Player A's tokens first and then Player B's into the same thin pool, B always gets the worse exit on any token both hold. Deterministic order is a built-in unfairness; random order is a lottery.	Valid	Accept. Change valuation to independent simulated liquidation : value each player's positions as if sold alone against the current reserves, without mutating the pool. Symmetric, price impact still counts, no order dependency. Tokens are throwaway, so no actual sale is needed.	
VP-6	Shared thin pools make the game about the opponent's pool moves, not markets. With two players in a small pool, pumping a token the opponent holds and dumping before the buzzer is the dominant strategy. That is PvP grieving, not trading.	Partially valid	Accept for the capstone as a known property (mutual price impact is stated as a feature in the PRD). Mitigate: seed pools deep relative to bankroll (reserve $\geq 50 \times B$) so a single bankroll cannot move price more than a few percent. Document the ratio as a config constant.	
VP-7	Two revenue streams is thin math. At the E=\$25 tier the draft's own numbers give ~\$3.90 per duel and need 500 duels per day for ~\$1.7k per day. A 1v1 format needs two matched humans per duel; matchmaking liquidity is the cold-start killer of every PvP product.	Valid	Accept. Remove the revenue claim from the capstone LOI or present it as illustrative only. Add matchmaking liquidity to the risk section. State that freeroll retention, not revenue, is the first metric.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
VP-8	The trustless claim does not hold for the stated mainnet v1. The deep dive's mainnet v1 uses Privy server wallets and operator-signed settlement. That is a trusted operator. Only the PDA-owned v2 is trustless.	Valid	Accept. Say "trustless in the capstone and in mainnet v2; mainnet v1 is operator-settled with published balances." Do not claim trustless for a design that is not.	
VP-9	Legal framing is weaker than presented. Entry fee plus a prize decided by a short-horizon price outcome looks like a wager on price movement. The DFS precedent rests on longer horizons and statistical skill evidence that does not yet exist here (see VP-2).	Valid	Accept. Keep freeroll and friendly modes as the only launch modes. Downgrade "DFS-style skill contest" from a defense to a hypothesis pending counsel and the backtest.	
VP-10	The streamer incentive is broken at these tiers. 1% of a \$50 pot is \$0.50 per duel. A streamer earning thousands per day in creator fees will not run a segment for that.	Valid	Accept. Reframe the streamer value as engagement and content (a live segment with a scoreboard), with the pot share as a bonus. Consider a per-duel flat host fee or sponsored purses. Record the 1% figure as a placeholder.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
VP-11	Market figures are unsourced or secondary. “\$1.6B streamer-token volume / 30d,” “110M social interactions,” “625k users,” “\$20M/week creator payouts” come from press and aggregator posts and were not independently verified.	Valid	Accept. Cite each figure with source and date in Part 1, or soften to “reported.” Verified 2026-09-09 from Crypto Briefing: fomo weekly revenue crossed \$2M; fomo briefly overtook Axiom in daily fees on 2026-08-06; pump.fun launched a social trading feature on 2026-08-07.	
VP-12	“Rug” undercuts the trust message. The product sells a fair referee and the name borrows the scam word. It also invites app-store and payment-processor friction later.	Partially valid	Reject as a blocker, accept as a known trade-off. Founder chose the name for stickiness and clip value on 2026-09-07. Record the trade-off; keep the legal entity name separate.	

B.1: Red team: Target Markets & User Profiles

#	Attack	Proposed validity	Proposed decision / change	Founder decision
TM-1	Segment 3 (spectators) depends on a third party in alpha. The spectator value needs Melee to exist, stay permissionless, and accept the Duel account as a resolution source. None of that is confirmed.	Valid	Accept. Keep spectators as a Phase 6 segment, not a v1 target. Add “confirm Melee permissionless creation and resolution rules” to the validation sprint (already day 6).	
TM-2	Segments 4 and 5 are admitted guesses. The draft says so itself. They pad the list.	Valid	Accept. Cut to three segments in Part 1 (retail traders, streamers, spectators) and move agents and sponsors to a “later segments” note.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
TM-3	User profile numbers are invented. “Bankroll \$100–\$5k,” “top-100 streamer,” “follows 5–10 traders” have no interview or data behind them.	Valid	Accept. Remove the numbers or label them assumptions. Replace with what the founder actually observes in the communities they are in.	
TM-4	A likely better channel is missing: trading terminals. Axiom, Photon, BullX, Trojan, and Padre own the trader’s screen. An embeddable duel widget or API inside a terminal reaches more players than any streamer. The draft mentions “be the API any terminal can embed” only as a fomo counter-move.	Valid	Accept. Add “trading terminals” as a distribution segment. Note that the API-first shape (permissionless instructions) already supports it with no program change.	
TM-5	Nobody has been asked whether they would pay entry. Every segment is untested against paid play. Freeroll demand and paid demand are different products.	Valid	Accept. State it. The first paid-mode test is gated on counsel anyway.	

B.1: Red team: Competitor Landscape

#	Attack	Proposed validity	Proposed decision / change	Founder decision
CL-1	Missed direct competitor: PerpClash. Live 1v1 timed trading duels on BTC, ETH, and SOL perps, up to 20x leverage, wallet sign-in with no deposit required, season points feeding a prize vault split among the top 100. Found 2026-09-09; the 2026-08-31 coverage check missed it. It is the closest shipped analog to the timed-duel format.	Valid	Accept. Add to the table as Direct. Verify whether positions are virtual (the “no deposit” line suggests so) and what “every trade is on-chain” means. Differentiation: Rug Royale has escrowed stakes and memecoin spot fills; PerpClash has majors, leverage, and a no-stake season model. Their season-points model is also a design to learn from for the Ladder feature.	
CL-2	Missed adjacent products in social trading. pump.fun launched a social trading feature on 2026-08-07. fomo has copy-trading and cross-chain. Vector (Tensor’s social trading app) and Kolscan (public KOL PnL leaderboards) are the “proof of PnL” incumbents. Sol-PvP and SOL Arena appear in search as Solana PvP games. None were checked.	Valid	Accept. Add a “social trading / PnL proof” row (pump.fun social, fomo, Vector, Kolscan) as substitutes for the status motive. Verify Sol-PvP and SOL Arena; expected to be chance games like DUELPVP.	
CL-3	SolDuel’s zero may indict the category, not the variant. The draft reads it as “escrow alone is not enough.” The other reading is “nobody wants 1v1 trading duels.” No evidence separates the two.	Valid	Accept. Present both readings. The validation sprint (streamer yeses, access DMs, manual duels) is the test that separates them.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
CL-4	PvPBattle’s traction is unknown, and if it has any, paper trading with stakes is enough for users. Then “real fills” is a differentiator for the LOI, not for the customer. Verified 2026-09-09: PvPBattle is custodial, settles virtual positions against Binance mark price, 30-second to 15-minute battles, up to 25x leverage, example pots 0.5–4 SOL, no public traction numbers.	Partially valid	Accept the uncertainty. Note that PvPBattle’s 30-second battles confirm the market skews toward fast, chance-heavy formats, which sharpens VP-2 and VP-9.	
CL-5	fomo can ship 1v1 in a week. 625k users, \$2M+ weekly revenue, already runs PnL competitions. The stated moat is speed plus streamer relationships the founder does not yet have.	Valid	Accept. Say it plainly in the risk section. The honest moat candidates are: the open, permissionless protocol shape (anyone can embed it), the settlement-as-oracle design, and agent entry. Speed is not a moat.	
CL-6	Melee listed as “partner” without contact. No conversation has happened. Listing a partner you have not spoken to is a red flag to a panel.	Valid	Accept. Relabel as “prospective venue.”	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
CL-7	DefiLlama and Colosseum checks still not run. The draft admits it.	Valid	Accept. Run a DefiLlama raises search (terms: trading game, PvP, social trading, prediction) and a Colosseum Copilot search (duel, trading competition, PnL battle) before the final. Colosseum Frontier 2026 winners list (26 winners, grand champion CrowdBrain) contains no trading-duel project on first pass.	

B.3: Red team: Founder-Market Fit

#	Attack	Proposed validity	Proposed decision / change	Founder decision
FM-1	The FMF section is empty. Placeholders cannot be red-teamed. Everything below is against the skeleton.	Valid	Accept. Founder writes the section in first person before the refined LOI.	
FM-2	The stated unfair advantage is irrelevant to the capstone and to first PMF. SolEnrich's wallet graph powers an integrity layer that is Phase 6. It does not help ship the program or find the first 100 duelists.	Valid	Accept. Reposition SolEnrich as evidence of shipping ability (a live, paid, onchain-data API) and Solana data fluency. Keep the integrity layer as a later advantage, one line.	

#	Attack	Proposed validity	Proposed decision / change	Founder decision
FM-3	Distribution is the stated hard part, and the founder lists no distribution assets. No streamer relationships, no audience, no community role. The plan depends entirely on streamers.	Valid	Accept. Founder must either name real relationships and channels, or replace the streamer wedge with one they can execute alone (terminal API embed, Farcaster or X freeroll nights, Turbin3 cohort as first duelists).	
FM-4	Solo builder plus an AMM is scope risk. The internal pool adds a second protocol (constant-product math, fee accounting, invariant tests) on top of escrow and settlement. <code>join_duel</code> alone initializes about 14 accounts. Anchor experience level is unstated.	Valid	Accept. State Anchor and Rust experience honestly. Keep the AMM minimal (no LP tokens, no fee tiers, fixed seed). Move account initialization out of <code>join_duel</code> (see GC-1).	
FM-5	“Active memecoin participant” is not a credential. It shows domain exposure, not the ability to build a game or run a competitive community.	Partially valid	Accept the framing change. Domain exposure is still real fit for user empathy. Pair it with specific observations (what call-outs look like in chats, what people argue about) and with the shipping record.	
FM-6	Motivation reads as opportunistic. The idea came out of a scored shortlist, not a personal itch. Panels can tell.	Partially valid	Accept only if true. If there is a real story (a bet that could not be settled, a stream moment), lead with it. If not, the honest framing is “I trade in these rooms and this is the format they are missing.”	

Phase 2: Adversarial Analysis & Granularity Check

Rule 1: Atomicity (one use case = one instruction handler)

#	Finding	Validity	Proposed decision / change	Founder decision
GC-1	<code>join_duel</code> is one status transition but initializes ~14 accounts (2 vault authorities, 8 player token accounts, 3 pools, 6 pool token accounts) and funds them. Transaction size and compute are tight, and a single failure reverts the join. Account initialization is not a state transition and does not belong in the transition instruction.	Valid	Split: <code>create_duel</code> initializes the 3 pools and the creator's vaults (creator pays rent; pools seeded at create). <code>join_duel</code> initializes only the opponent's vaults and flips <code>Open</code> → <code>Active</code> . Nine handlers total.	
GC-2	<code>settle</code> does valuation, prize transfer, rake transfer, and status flip in one handler. That is one transition (<code>Active</code> → <code>Settled</code>) and is correct. With VP-5's simulated-liquidation change it no longer mutates pools, which reduces accounts and compute.	Invalid as an attack	Keep as one handler.	
GC-3	The PRD's single <code>refund</code> covered two transitions (<code>Open</code> → <code>Cancelled</code> and <code>Active</code> → <code>Refunded</code>). Draft v1 already split it into <code>cancel_duel</code> and <code>emergency_refund</code> .	Valid (already applied)	Keep the split.	
GC-4	<code>close_duel</code> closes up to 16 accounts in one call. Not a state transition, but a cleanup. Compute is fine (close is cheap). One handler is acceptable; a per-account-group variant is available if limits bite.	Invalid as an attack	Keep. Note the fallback.	

#	Finding	Validity	Proposed decision / change	Founder decision
GC-5	Freeroll mode with <code>entry_lamports == 0</code> has no prize path. The LOI mentions sponsored prizes; no instruction deposits one.	Valid	Add <code>sponsor_prize:</code> any signer deposits lamports into <code>Escrow</code> while <code>status == Open</code> . Emits <code>PrizeSponsored</code> . Ten handlers total. Or scope sponsored prizes out of the capstone explicitly.	

Rule 2: State ownership (every state change in a program-owned account or an explicit indexer)

#	Finding	Validity	Proposed decision / change	Founder decision
GC-6	<code>winner</code> , <code>final_a</code> , <code>final_b</code> are recorded on the <code>Duel</code> account. Correct. But the draft calls the <code>DuelSettled event</code> the resolution oracle. Anchor events live in transaction logs, which RPCs can truncate and which are not readable by other programs.	Valid	Change the claim: the <code>Duel</code> account state is the oracle; the event is a notification for indexers. Keep <code>Duel</code> open (do not close) for a configurable retention period, or make <code>close_duel</code> write a compact <code>DuelResult</code> receipt PDA first.	
GC-7	Bankroll B is program-minted play money in the capstone (program PDA is mint authority of a devnet quote mint). Only E is the player's money. The draft does not say this.	Valid	State it explicitly in the LOI and the demo. On mainnet B is the player's own deposit.	
GC-8	Streamer pot share (Phase 6) is described as an off-chain host link. Any value routing that is off-chain is a trust assumption.	Valid	Note for Phase 6: <code>host:</code> <code>Option<Pubkey></code> and <code>host_bps</code> on the <code>Duel</code> account, paid inside <code>settle</code> . No capstone change.	
GC-9	Lobby listing uses <code>getProgramAccounts</code> . Fine on devnet; on mainnet needs an indexer.	Valid, minor	Note it as client-side with a mainnet indexer dependency.	

Rule 3: Real signers (explicit, no hidden backend)

#	Finding	Validity	Proposed decision / change	Founder decision
GC-10	<code>emergency_refund</code> requires the admin. Since <code>settle</code> is permissionless and deterministic, the only way a duel gets stuck is a program bug. An admin-only escape hatch is a hidden trust assumption and a grieving surface (admin declines to sign).	Valid	Make <code>emergency_refund</code> permissionless: any signer may call it when <code>status == Active</code> and <code>now > end_ts + grace_secs</code> . Admin keeps only <code>init_config</code> and a future <code>update_config</code> .	
GC-11	<code>cancel_duel</code> says “Creator, or any signer with refund routed to Creator.” Pick one.	Valid	Any signer; funds always go to <code>creator</code> . Matches the permissionless pattern.	
GC-12	<code>settle</code> is signed by “anyone.” The <code>start = now + 60</code> delay and window bounds rely on the Clock sysvar, which is fine. But if nobody cranks <code>settle</code> , the duel sits Active. There is no backend and no incentive.	Partially valid	Accept: add a small settler tip (fixed lamports from rake or a config constant) paid to the <code>settle</code> signer, or accept that the players themselves crank it (the frontend shows a Settle button). Capstone: players crank.	
GC-13	Who is the mint authority for the throwaway mints and the quote mint in the demo? If it is the demo script’s keypair, the demo has a hidden backend that can mint.	Valid	Quote mint authority = program PDA <code>["mint_authority"]</code> . Throwaway token mints: authority burned after the demo script seeds them, or also a program PDA. State it.	

Rule 4: On-chain vs. client-side separation

#	Finding	Validity	Proposed decision / change	Founder decision
GC-14	PnL bars, timers, result cards, lobby, allowlist curation, host links, collusion detection are all listed as client-side. Correct.	Invalid as an attack	Keep the list.	
GC-15	“Token allowlist” is client-side on mainnet, but the on-chain <code>token_list</code> is chosen by the creator. Nothing on-chain stops a creator from listing a mint they control. On the capstone’s internal AMM this does not matter; on mainnet it does.	Valid, Phase 6	Note for mainnet: an on-chain allowlist account maintained by admin or a Merkle root, checked in <code>create_duel</code> . No capstone change.	
GC-16	Tier constants and pool seed ratios are described as “config constants” in prose. They must live in <code>Config</code> on-chain or be compile-time constants; otherwise they are a hidden backend parameter.	Valid	Put tiers: <code>[Tier; 3]</code> (bankroll per tier) and <code>pool_seed_ratio</code> in <code>Config</code> .	

Consolidated changes to carry into the refined LOI (pending founder confirmation)

Phase 1 1. Lead the value prop with the game and stakes; verifiability becomes the “why onchain” enabler (VP-1). 2. Name the skill-at-short-horizon question as the top unvalidated assumption; default to 15/60-minute windows; run or schedule the backtest (VP-2, VP-9). 3. State that the capstone proves escrow and settlement on a deterministic test market, not real-market fills (VP-4, GC-7). 4. Remove revenue projections from the capstone LOI; add matchmaking liquidity and fomo fast-follow to risks (VP-7, CL-5). 5. Trustless claim limited to capstone and mainnet v2 (VP-8). 6. Streamer incentive reframed; 1% becomes a placeholder (VP-10). 7. Cite or soften every market figure (VP-11). 8. Segments cut to three plus trading terminals; numbers in profiles removed or labeled (TM-2, TM-3, TM-4). 9. Competitor table adds PerpClash (direct), social-trading/PnL-proof row, Sol-PvP and SOL Arena (verify); Melee relabeled prospective venue; both readings of SolDuel’s zero (CL-1 to CL-6). 10. FMF written by the founder; SolEnrich repositioned as shipping evidence; distribution assets named or wedge changed (FM-1 to FM-6).

Phase 2 11. Handlers: `init_config`, `create_duel` (inits pools + creator vaults), `join_duel` (inits opponent vaults, flips Active), `sponsor_prize`, `swap`, `settle`, `cancel_duel`, `emergency_refund` (permissionless), `close_duel`. Ten if `sponsor_prize` stays. 12. Settlement valuation = independent simulated liquidation, no pool mutation, no order dependency (VP-5). 13. `Duel` account state is the oracle; event is notification; add `DuelResult` receipt or retention rule (GC-6). 14. Tiers and pool seed ratio stored in `Config`; quote mint authority is a program PDA (GC-13, GC-16). 15. Phase 6 notes: on-chain host share, on-chain allowlist, settler tip, mainnet indexer (GC-8, GC-12, GC-15).

Dialogue rounds (founder ↔ AI, verbatim)

The tables above are the AI's opening attack list. The rounds below are the actual red team exercise: founder text, AI attack, founder rebuttal, AI counter, decision.

Round 1: Project overview (2026-09-09)

Founder's original text (unedited): > Rug Royale is a 1v1 PvP trading-duel protocol on Solana. Two traders lock an entry into a trustless escrow, each trades an equal, fixed bankroll on an agreed token list for a fixed window (5min, 15min, 30min, etc.), and the program settles the winner from verifiable onchain fills. PnL flexing in memecoin trading today runs on screenshots that nobody can verify, settle, or monetize; Rug Royale makes the flex settleable. The capstone ships the escrow, swap, and settlement program on devnet with a thin duel-room frontend. Post-capstone, the same settlement state becomes a public resolution oracle for spectator markets and generalizes to N-player battle-royale tournaments.

Founder note: this text was AI-compiled from prior iterated research; the founder adopted it as the starting draft.

#	AI attack	Founder rebuttal (verbatim)	AI counter	Decision
R1-1	The bankroll is unexplained. "Lock an entry" and "trade a fixed bankroll" are two pots; the reader cannot tell what is at risk or what the winner gets. In the capstone the bankroll is program-minted, which makes the omission worse.	"This could be updated in several ways. I would think the traders both put up \$25 and trade, or the winner gets the other persons stake. I think we need to flesh this out."	The two options are one design decision already made in the PRD: (A) trade your own stake, winner takes the loser's balance; (B) entry escrowed, identical separate bankroll, winner takes the pot minus rake. B bounds loss and keeps PnL% comparable, and is the only honest option on devnet where the bankroll is play money.	Valid. Keep design B. Add one sentence to the overview stating entry vs. bankroll and what the winner receives.

#	AI attack	Founder rebuttal (verbatim)	AI counter	Decision
R1-2	“Verifiable onchain fills” overclaims what the capstone ships: fills are against a program-owned pool on throwaway mints.	“Ideally we would be shipping against real memecoin markets. Hard to bridge this gap on devnet, without mocking it up in some way, or just testing in prod.”	The fix is wording, not engineering: “settles from onchain state,” capstone = deterministic internal market, mainnet = Jupiter. Option to investigate: Surfpool mainnet-fork local validator to demo against real pools without prod.	Valid. Reword. Add “investigate Surfpool mainnet fork” to the PRD Phase 0 checklist as an option.
R1-3	The problem statement leads with verification, which is not the felt pain; it mixes trader, streamer, and protocol motives in one clause.	“The problem statement can be readjusted that there isn’t a way to easily trade against opponents, settle who is a better trader, gain status, and turn their skill into a spectacle, verified for clout, with rewards on the table.”	The reframe is correct: the pain is no way to play against a specific opponent for money and status; verification is what makes the status claim stick. Six verbs is too many for the overview; keep “settle who is better” and “win money” there, the rest in the value prop.	Valid. Rewrite the problem sentence around opponent, status, and rewards; verification demoted to enabler.
R1-4	No user appears in the paragraph. “Two traders” is a role, not a market.	“The app is really for people who want to go beyond just trading memecoins for profit, and allows you to monetize a new aspect of crypto, applying the trading game to a more explicit game you can actually play against another person to win money, in addition to profits.”	That is a motive, not a person. Name them: memecoin traders already active in pump.fun and X trading communities. “Monetize a new aspect of crypto” is too abstract for a panel.	Partially valid on the founder’s side; attack held. Name the user concretely in the overview.

#	AI attack	Founder rebuttal (verbatim)	AI counter	Decision
R1-5	The window list leads with the weakest case (5 min) and ends with “etc.”; 30 min is unreconciled with the PRD’s 300/900/3600.	“I’m open to considering timeframes here. I think 5 minute intervals is good to test out, the best is likely 10 I would imagine, but I don’t know what users want to do until we test it.”	Correct answer, and it changes the design: overview says “short fixed windows” only; PRD makes <code>window_secs</code> a configurable value in <code>Config</code> rather than a hardcoded set, so 5/10/15 can be tested without redeploy. Skill and legal concern at very short windows stays open pending the backtest.	Valid. Overview: “short fixed windows.” PRD change: configurable window set in Config.
R1-6	The last sentence names two future products (oracle, battle royale); reads as scope creep.	“I do think the product could be expanded to have team battles, players vs. agents, and other fun games that take this principle and scale it.”	Vision accepted; point about the paragraph held. One clause covers all of it: “the same escrow-and-settle core extends to team battles, tournaments, and agent opponents.”	Partially valid. Keep the vision as one clause; drop the named oracle mechanism from the overview.

Founder confirmations (2026-09-10, verbatim): - R1-1: “I like option B for A1-1.” → Design B confirmed: entry escrowed, identical separate bankroll, winner takes the pot minus rake. - R1-2: “Worth diving deeper for testing in prod for R1-2.” → Founder wants to explore demoing against real markets. AI note: investigate Surfpool mainnet fork first (no real funds); a mainnet test with real memecoins is a post-capstone option because it requires the Jupiter path and real money. - R1-4: Attack accepted. Founder’s user definition: “the users are memecoins traders on pump.fun and fomo, who already follow KOL callouts, like placing their own trades, and earning money from good plays, winning with their calls as opposed to other communities. Traders who like to flex with PnL screenshots, and retail who want a chance to bet on the action or do a low stakes game like call of duty, to make trading more fun.” → Two named users: (1) pump.fun / fomo traders who follow KOL call-outs, place their own trades, and flex PnL; (2) retail who want to bet on the action or play a low-stakes competitive game. The Call of Duty comparison (casual competitive matches) carries into the value prop. - R1-6: “I like the scope change for R1-6.” → One-clause vision confirmed.

Assembled overview after Round 1 (confirmed 2026-09-10, later amended by Round 2a): > Rug Royale is a 1v1 trading-duel protocol on Solana for pump.fun and fomo traders who follow KOL call-outs, place their own trades, and flex their PnL, and for retail who want a low-stakes competitive game on top of the action. Today there is no way to challenge a specific opponent, settle who is the better trader, and win money and status doing it; screenshots cannot be verified and call-outs cannot be settled. Two traders each escrow an entry, trade an identical bankroll on an agreed token list for a short fixed window, and the program pays the pot

minus rake to the higher PnL%, with every fill and the result onchain so no referee is needed. The capstone ships the escrow, swap, and settlement program on devnet against a deterministic internal market, with a thin duel-room frontend. The same escrow-and-settle core extends to team battles, tournaments, and agent opponents.

Resulting changes carried forward: overview rewritten with the named users (see refined LOI); PRD: `window_secs` configurable in `Config`; PRD Phase 0: investigate Surfpool mainnet fork for the demo; post-capstone option: mainnet test against real markets.

Round 2: Value proposition (opened 2026-09-10, founder text on file; rebuttals pending)

Founder's original text (unedited): > Rug Royale turns the most common social behavior in memecoin trading, proving you are the better trader, and turns it into a game with real stakes and trustless settlement. An early mover in the "entertainmentFi" space, Rug Royale appeals to users of pump.fun and fomo, who already trade socially, and enjoy the game of trading. Two players lock an entry (E) into an escrow and trade an identical bankroll (B) for 5, 10, 15, or 30 minutes, at the buzzer, positions are force sold to the quote asset, and the higher PnL% takes the prize (2E minus rake). Separating E from B bounds each payers loss, keeps PnL% comparable across players, and creates two revenue streams: a rake on the price and integrator fees on every swap. Becuase escorw, fills, and settlement all happen onchain, results cannot be fakes, and disputes on faking PNL cannot arise, as there is no bookmaker or broker needed. In web2 this product can't exist without a borkerage and gaming license. On solana, it is a (relatively) simple anchor program plus ready made DEX infra. OUR tagline: "Stop flexing fake screenshots. Settle it onchain" Why now: 1) Social trading on pump.fun and fomo is a measurable behavior, streamer tokens did about \$1.6B vol. over 30 days, and pump.fun pays roughly \$20M per week in creator fees. Streamers need new interactive segments, they could accept challenges from each other, or from their audience. 2) fomo runs sponsored many-player PNL competitions and reports 110M in social interactions. Demand for competitive trading is proven, a head-to-head, stake your own format is not served. 3) Melee Markets is an emerging permissionless prediction market on solana. It can open up another side of the market where spectators can have an active "who wins?" prediction market, so the duel itself is also content, without the protocol having to run a book.

AI attacks (2026-09-10, verbatim as issued) and founder rebuttals (2026-09-11, verbatim):

Attack 1. The best idea you have is missing. In the team formation post, the line that worked was "money is following the path news took online and becoming entertainment." Here entertainmentFi appears only as "an early mover in the space," which is a position claim, not a value. And the Call of Duty shape we agreed belonged here, a ten-minute competitive match you can play between trades, is absent. The paragraph explains mechanics well and never says what it feels like to play. > **Founder:** "the product takes an activity that already is treated like a game by many, trading, and makes it an explicit and profitable game for participants. Think Call of Duty for trading, with intense, short, and repeateable matches where you can prove yourself in real time. We are looking as well to target stonkfun ecosystem coins as well, and this creates real incentives around different paired communities, coupled with activity that results in drips for participants, making a multi-sided model." > **AI counter:** Accepted; this text becomes the spine

of the value prop. The news-to-entertainment thesis is still absent; add one sentence of setup. > **Decision:** Valid. Rewrite around “Call of Duty for trading” + StonkFun multi-sided incentive + one thesis sentence.

Attack 2. The revenue sentence describes a product you are not building yet.

“Integrator fees on every swap” requires Jupiter, which is mainnet Phase 6. The capstone’s internal AMM earns nothing. Freeroll mode has no rake. So on demo day both streams are zero, and by your own earlier math the mainnet version earns about \$3.90 per duel and needs 500 duels a day to matter. A panel will do that arithmetic. Also “rake on the price” should read prize. > **Founder:** “I think I need a little help fleshing out the revenue economics of the platform. I had thought a 1% fee on the pools goes back into the platform, but I also think there could be a novel mechanism of our own token (paired with stonk) that participants get rewarded in, and this also drips stonk to all eco participants for activity. Could use some help with this flywheel.” > **AI counter:** Three things separated. (1) Capstone revenue: rake on the pot (0 in freeroll, configurable) + the internal AMM swap fee, which the protocol sets because it owns the pool. (2) Mainnet: Raydium CPMM fees go to LPs and Raydium, not to the protocol; swap revenue comes only via a Jupiter integrator fee or a frontend fee. So mainnet = rake + integrator fee, ~\$3.90/duel at the E=\$25 tier. (3) Flywheel: launch a Rug Royale coin on StonkFun in reward mode paired with STONK; 3% transfer tax drips STONK to holders; rake buys the coin; duelists who hold it earn STONK drip from platform activity. Coherent loop (duels → StonkFun swap volume → STONK burn → rake buys coin → coin drips STONK to duelists), but it is a token launch: regulatory and scope expansion the capstone should not carry. > **Decision:** Partially valid on the founder side. Value prop states rake + swap fee and the multi-sided incentive without a token. The own-coin flywheel goes in a one-sentence “later economics” note, clearly post-capstone. Correct “1% of pool fees” misconception.

Attack 3. Three overclaims in one sentence. “Results cannot be faked” holds. “Disputes cannot arise” does not: players will dispute pool manipulation, a stuck settle, a token that went to zero mid-window, or the valuation method. “No bookmaker or broker needed” is true for the capstone and the PDA-owned mainnet design, but the deep dive’s mainnet v1 uses operator-signed settlement through Privy wallets. Either commit to the PDA design for mainnet or narrow the claim. > **Founder:** “Yea this is loose language we should adjust. Of course people will complain if they lose. I think we can just narrow the claim here.” > **Decision:** Valid. Narrowed claim: results cannot be faked and settlement needs no operator; rule and manipulation disputes are handled by published rules and the integrity module.

Attack 4. Every why-now number is unsourced. \$1.6B streamer-token volume, \$20M per week creator fees, 110M social interactions. These came from press and aggregator posts and none were independently checked. One verified figure exists from Crypto Briefing: fomo weekly revenue crossed \$2M and it briefly overtook Axiom in daily fees on 2026-08-06. Use what you can cite, or write “reported.” > **Founder:** “I’ll need to flesh this out with the actual sources, ditto for the stonkfun stuff.” > **Decision:** Valid. Founder sources every figure before the final. Verified and available now: fomo weekly revenue > \$2M (Crypto Briefing, Aug 2026); STONK ~\$140M market cap on ~\$135M daily volume after the Raydium LaunchLab integration (The Block, 2026-09-06).

Attack 5. The paragraph contradicts two Round 1 decisions. You listed 5, 10, 15, or 30 minutes; Round 1 settled on “short fixed windows” with the values configurable. And “positions

are force sold to the quote asset” is the settlement method the opening list flagged as order-dependent and exploitable in a shared pool, with simulated independent liquidation proposed instead. That decision is still yours to make, but the value prop should not lock in a method the requirements section may change. > **Founder:** “We should adjust this to be short fixed windows so we aren’t locked into a decision. I agree with this pushback, well done.” > **Decision:** Valid. “Short fixed windows”; settlement method described generically (“positions valued onchain at the buzzer”) pending the Phase 2 decision on simulated liquidation.

Attack 6. Two lines invite the wrong questions. “In Web2 this can’t exist without a brokerage and gaming license” tells a regulator you think it is gaming. Say licensed custody and leave gaming out, or address it head on in the risks. And Melee as why-now number three is a vendor dependency, not a market signal. Melee is in alpha and you have not spoken to them. It belongs in the growth section as a prospective venue, not as a reason the market is ready. > **Founder:** “Let’s just say licensed custody, agreed, and also agreed to put melee in the growth section, likely not a part of the product from day 1 as that is still launch dependent.” > **Decision:** Valid. “Licensed custody”; Melee moves to growth as a prospective venue.

Positive noted: “streamers could accept challenges from each other or from their audience” is stronger than the 1%-of-pot streamer story. Carry into GTM.

Resulting changes carried forward: value prop rewritten (see refined LOI); economics: capstone = rake + internal AMM fee, mainnet = rake + integrator fee; own-coin-on-StonkFun flywheel logged as a post-capstone option; all figures to be sourced by founder.

Round 2a: Founder-initiated scope change: bet on the StonkFun ecosystem (2026-09-11)

Founder’s proposal (verbatim): > Stonkfun.xyz is really big right now, and unlocking new primitives on SOL in the RWA space. Because our tooling is agnostic, I think what we do is make a specific bet on the stonk eco. I think this makes it more interesting than an “any token” duel, we let players pick a StonkFun coin or an xStocks pair, and keeps money continually flowing into the Stonk ecosystem. StonkFun launches run on Raydium LaunchLab and graduate into CPMM pools, we would have plenty of onchain data to settle the duels in a straightforward way. We can even use sources like stonk.fyi to scrape data, or build our own pipeline from the stonkfun Api.

Facts verified 2026-09-11: - StonkFun (stonkfun.xyz) is a Solana launchpad where each new coin is paired with a quote asset of the creator’s choice: xStocks tokenized equities from Backed Finance (SPYx, NVDAx, QQQx), PreStocks pre-IPO tokens (OPENAI, ANTHROPIC), crypto (ZEC, WBTC, HYPE, xSOL), plain SOL, or another StonkFun coin. Native token STONK; a share of platform fees buys back and burns STONK. - Raydium LaunchLab integration announced ~2026-09-05/06 (first partner after LaunchLab opened “any token pair” on 2026-09-07). STONK rose ~250% in a day to ~\$140M market cap on ~\$135M daily volume (The Block, 2026-09-06). stonk.fyi (unofficial dashboard) shows ~\$7.62M fees this week, 1.3K tokens, 55.2K holders, 14.48% of supply burned. - Mechanics: new launches trade on a LaunchLab bonding curve, then graduate into a Raydium CPMM pool once the curve raises its target in the quote asset. CLMM is used for direct launches and for STONK itself. - Data: free public API at `/api/public/v1`, no key, 300 req/min per IP; tokens (market data, launch record, market cap, volume, burns), graduation status (`new` / `aboutToGraduate` / `graduated`), Raydium pair and pool

details, launch ledger. Bitquery exposes trades, prices, candles, holders. The API has no competitions, leaderboards, PnL, or duels. - xStocks: Backed Finance, 60+ US equities, 1:1 backed under Swiss oversight, 24/7 on Solana. Backed blocks US persons; trading xStocks on a DEX from a US IP is not compliant with the issuer's terms. KYC at primary mint only.

AI assessment: | # | Point | Type | Note | |—|—|—|—| | S-1 | Sharper wedge than “any memecoin.” A named community (55K STONK holders, launch creators), a partner with an aligned incentive (duel swap volume → StonkFun fees → STONK buyback), and a narrative that is literally the entertainmentFi thesis (memecoins paired with stocks). | Strength | Adopt as the launch wedge. | | S-2 | Timing is real but young. The ecosystem is one week into a 250% move. Launchpad tokens mean-revert. If StonkFun fades before demo day, a StonkFun-only product has a dead wedge. | Risk | Keep the program token-agnostic (it already is: `token_list` is any mints). The bet lives in the allowlist, branding, and GTM, not in the program. Wedge, not dependency. | | S-3 | Regulatory surface grows. xStocks are tokenized securities; the issuer blocks US persons. A wagered contest that routes players into xStocks pairs from the US stacks a securities question on top of the gaming question. | Risk (top) | Launch allowlist = StonkFun coins quoted in SOL or STONK. xStocks-quoted duels gated on geo-blocking and counsel. Freeroll for the capstone as before. | | S-4 | PnL% comparability across quote assets. A NVDAx-paired coin and a SOL-paired coin have PnL% in different units. | Design | Already solved by the per-duel token list: both players trade the same list, so the quote is shared. Add: one quote asset per duel; bankroll denominated in it. | | S-5 | Thin, fresh CPMM pools raise the pre-hold-and-pump risk and price impact from a single bankroll. | Risk | Allowlist = `graduated` coins with a pool-reserve floor read from the API/pool accounts. Cap bankroll tier relative to reserves. | | S-6 | Settlement data path is clean. Graduated coins sit in Raydium CPMM pools whose reserves are onchain; the program can value positions against pool state or Jupiter fills. This also makes the Surfpool mainnet-fork demo more attractive: fork mainnet, trade real StonkFun pools locally. | Strength | PRD Phase 0: evaluate Surfpool fork with real StonkFun CPMM pools as the stretch demo. Internal AMM stays as the deterministic fallback. | | S-7 | Fast-follow risk shifts from fomo to StonkFun itself (they have “reward modes” and creator tooling). Launchpads rarely build games; featuring a third-party arena is more likely. | Risk | Pitch StonkFun early on the volume → burn flywheel; ask for a feature slot, not permission. | | S-8 | LOI changes if adopted: overview and value prop gain a StonkFun sentence; segments add STONK community and StonkFun launch creators; competitors add “none StonkFun-native” (verified: API has no games); risks add ecosystem concentration and securities surface; GTM adds StonkFun partnership. Program: no change. | Scope | Founder decides framing before Round 2 rebuttals so the value prop is written once. |

Founder decision (2026-09-11, verbatim): “I think instead of xStock pairs, we basically just focus on stonkfun.xyz tokens that have a lot of momentum on the platform. I think that builds us in a user base, and has interesting potentiality with their yield and drip mechanics, funneling buys and rakes into their eco, etc.”

→ **Adopted:** StonkFun momentum coins are the launch allowlist. No xStocks pairs. Program stays token-agnostic. S-3 (securities surface) is removed from the launch path by this decision; it returns only if xStocks-quoted duels are ever added.

Follow-up fact (verified 2026-09-11): StonkFun “reward mode” coins carry a 1% or 3% transfer tax on every transfer (Token-2022 transfer fee, 100 or 300 bps), redistributed to holders

in the coin's quote asset (v3: 3% transfer tax; earlier launches: 85% of a 4% pool fee). Standard-mode coins have no tax. Launch shape: 1,000,000,000 supply, 6 decimals, bonding curve sells 79.31% of supply, graduation target in the quote asset.

#	Point	Type	Note
S-9	Reward-mode coins tax every transfer 1–3%. In a 15-minute duel with many swaps, tax drains bankroll and can dominate PnL%. The game becomes “who swaps least.”	Design	Launch allowlist = standard-mode graduated coins, or reward-mode coins with a per-duel swap cap. Decide per tier.
S-10	The vault holding a reward-mode coin during a duel accrues holder rewards in the quote asset (the drip). Settlement must count rewards received by the vault as part of final value, or the player who held longer is under-counted.	Design	Value = quote balance + simulated liquidation of positions + rewards claimed to the vault before valuation. Adds a claim step to <code>settle</code> for reward-mode coins.
S-11	Token-2022 transfer-fee tokens require <code>transfer_checked</code> with the fee extension and Token-2022 ATAs for vaults and pools. The capstone's internal AMM and vaults must support Token-2022 or the mainnet path will not.	Program	PRD change: vaults and pools use the Token-2022 interface (<code>anchor_spl::token_interface</code>) so a transfer-fee mint works without a rewrite. Capstone demo mints can be Token-2022 with a 3% fee to prove it.
S-12	“Funnel buys and rakes into the eco”: duel swap volume creates StonkFun and Raydium fees (STONK buyback), and the protocol rake could be taken in STONK or routed to a STONK burn. That is a concrete partnership offer.	GTM	Add to the value prop and GTM as the StonkFun pitch. Rake destination is a config field; no program logic change.

Round 3: Target markets & user profiles (2026-09-11)

Founder's original text (unedited; Draft v1 section plus a new StonkFun segment on top): > **Target market segments (2–5).** > Stonkfun eco participants who want to trade between their community coins within the eco, or dface off against traders of other ecos. Stonk side has a beneficial piece where the transfer fees let a rising tide raise all boats. > Solana memecoin retail traders who trade daily on pump.fun, Axiom, Photon, or BullX and post PnL. Largest segment; primary players. > pump.fun and X livestreamers who monetize through

creator fees and need interactive formats. Distribution channel and secondary revenue recipients. > Trading-content spectators who watch trench streams and bet on outcomes elsewhere (Polymarket, Pumpcade). Audience and, via Melee, the “handle.” > Trading-agent builders (Solana Agent Kit, elizaOS, Colosseum teams) who want a fair public arena for bots. Phase 2. > Sponsors and token teams who fund competitions as marketing (as fomo’s sponsors do today). Freeroll prize funders. > > **User profiles (top 3).** > | Profile | Who | Job to be done | Today’s workaround | Why Rug Royale | > |—|—|—|—|—| > | The on-stream degen | Trades memecoins daily, active in stream chats, posts PnL screenshots, bankroll \$100–\$5k | Prove I am the better trader and win money doing it | Screenshots, call-outs in chat, unverifiable bets between friends | Onchain proof, bounded stake, instant settlement, shareable card | > | The streamer / host | Top-100 pump.fun or X trench streamer, earns creator fees, needs segments that hold viewers | Keep chat engaged, add a revenue line | Bounties, giveaways, sponsored comps | Host link routes 1% of every pot to the host; live PnL overlay for OBS | > | The spectator | Watches trench streams nightly, follows 5–10 traders, bets on Polymarket occasionally | Have a rooting interest in a live event | Watch and comment; no stake | Live PnL bars; later, a “who wins?” market on Melee resolved by the duel’s own settlement |

Attack 1. The first retained cohort is not in the profile table. The confirmed PMF says the first cohort is StonkFun traders who call each other out over a launch. None of the three profiles is that person. The on-stream degen is the pre-pivot user. The top profile has to be the StonkFun call-out trader, or the PMF and the profiles describe two different products. >

Founder: “We need to add in the stonkFun call-out trader, and adjust accordingly.” >

Decision: Valid. StonkFun call-out trader becomes profile 1.

Attack 2. The StonkFun segment hides two products and one overclaim. “Trade between their community coins” and “face off against traders of other ecos” are different games. Two holders dueling on the same coin fits the design. A coin A holder versus a coin B holder does not, because the duel requires both players on the same coin so PnL% is comparable; an inter-community duel needs a neutral coin or a champion format. Decide which one is v1. And “transfer fees let a rising tide raise all boats” is backwards for duelists: the 1 to 3% tax drips to holders of that coin and drains the two people swapping. It is a benefit to the community, a cost to the players. > **Founder:** “Let’s adjust to make this segment more clear.” > **AI counter:** Clarity requires the decision. Recommendation: v1 = same-coin duels (two players, one agreed StonkFun coin). Cross-community “champion” duels on a neutral coin are a later format. Rising-tide claim rewritten as: the community benefits from duel volume (fees, drip to holders); duelists pay the tax, so v1 favors standard-mode coins. > **Founder (second pass):** “So I’m of two minds, a duel between the same coin is easier and potentially interesting, however, if both traders just sit in the same coin, it isn’t exactly that engaging. Also, stonkfun encourages holding, so it might not be the best thing on the same coin. Rival coins within stonkfun is likely a more engaging feature, especially if they are both paired to Stonk, the trades both benefit the broader eco anyway.” > **AI counter (second pass):** Same-coin with buy-and-hold ties (identical PnL%); reward-mode tax punishes the active trading it needs. Rival coins with a shared quote (STONK) keep PnL% comparable, give a community-vs-community narrative, benefit both communities, and remove the shared-pool settlement-order exploit (two coins, two pools). Named risk: if buy-and-hold dominates, this is SolDuel’s “champion token” model (zero users); mitigations are open trading through the window, short windows, and unranked/freeroll launch modes. Program resolution: replace the shared token list with one coin per player plus one shared quote asset; same-coin is the special case where both pick the same coin; format is a

client-side choice. > **Decision (confirmed 2026-09-11):** Valid. Program: per-player coin, shared quote asset. Launch narrative: rival StonkFun coins paired to STONK. Same-coin offered as an option. Carry to Phase 2 requirements.

Attack 3. Six segments, and the assignment caps at five. Agents and sponsors were already flagged as guesses in the opening list, and you agreed. With the pivot, “sponsors and token teams” has a concrete replacement: StonkFun coin creators and launch teams, who want volume on their coin and can fund a freeroll. Agents move to a “later” note. > **Founder:** “we move agents to later, and add stonkfun coin creators and launch teams who want to build buzz and trade their product.” > **Decision:** Valid. Agents → later note. New segment: StonkFun coin creators and launch teams.

Attack 4. “Largest segment; primary players” now points at the wrong group. After the pivot, primary players are StonkFun traders. The pump.fun, Axiom, Photon, and BullX crowd is where they come from and the expansion pool. Label it that way or the segment list argues with the overview. > **Founder:** “We need to adjust to our new segment so it lines up with our other changes.” > **Decision:** Valid. Memecoin retail relabeled as origin pool and expansion segment.

Attack 5. The streamer profile contradicts two decisions. “Host link routes 1% of every pot” was downgraded to a placeholder in Round 2, and the stronger streamer story from your own rebuttal, streamers accepting challenges from peers or their audience, is missing. The job-to-be-done should read as content and engagement, with any pot share as a bonus. > **Founder:** “Streamers were a good potential idea, but as a first profile, reads more as content and not the onchain piece. This should be a later expansion if we prove PMF.” > **Decision:** Valid, and the founder went further: streamers move out of the top profiles to a later segment, gated on PMF.

Attack 6. The invented numbers are still there. Bankroll \$100 to \$5k, top-100 streamer, follows 5 to 10 traders. You accepted this attack in the opening list. Replace with what you actually observe or delete. > **Founder:** “We need to sharpen these numbers and make realistic with our pivot.” > **AI counter:** Numbers in profiles must be observed or sourced. Available sourced ecosystem figures: ~55.2K STONK holders and ~1.3K StonkFun tokens (stonk.fyi, 2026-09-11). Profile-level figures (bankroll, follows) should be described qualitatively unless the founder has observed them. > **Decision:** Valid. Invented figures removed; ecosystem figures cited; qualitative descriptors otherwise.

Attack 7. The demand coverage check is missing. The assignment requires a coverage check against real ecosystem demand for this section. Nothing in the paste addresses evidence per segment or what is untested. The spectator segment also still leans on Melee as “the handle,” which Round 2 moved to growth as a prospective venue; keep spectators as a later segment. > **Founder:** “Can we review this again before I comment.” > **Founder (after review):** “because these products are all permissionless, I don’t really have someone to ask atm. I think we need to just make a measured bet, and assume that they might want this feature.” > **AI counter:** Partially valid. Protocols cannot be asked; the StonkFun team, coin creators, and players can. “No one to ask” is “not asked yet.” Accepted for the LOI as a measured bet: each row states the assumption, the evidence, and the cheapest test, and marks whether the test was run before submission. > **Decision:** Valid with the measured-bet framing. Demand check drafted by AI from founder decisions; confirmed with the assembled section.

Resulting changes carried forward: segments cut to five with StonkFun call-out traders first and coin creators added; agents and streamers moved to later notes; profiles led by the StonkFun call-out trader; invented figures removed; demand check written as measured bets; Phase 2: per-player coin with shared quote asset.

Round 4 (prep): Competitor coverage sweep, AI-run (2026-09-11)

Run before the founder’s competitor list so the round can attack a complete table. Queries: DefiLlama raises for social-trading / trading-game funding; StonkFun-adjacent games or leaderboards; Solana “trading battle / PnL battle / trade duel” launches in 2026; plus verification searches for each new name.

Name	Type	What it is (verified)	Relevance	Status
TradeFighters (tradefighters.fun)	Direct, unverified	Search description: “challenge traders in real-time duels, compete to prove your trading skills, and climb the leaderboard.” Site served only a logo on fetch 2026-09-11.	Same pitch as Rug Royale by description. Must be verified before the final (X account, launch status, real or paper trades, stakes).	Verify
PerpClash (perpclash.com)	Direct	1v1 timed duels on BTC/ETH/SOL perps, up to 20x, wallet sign-in, no deposit, season points → prize vault split among top 100. Root URL 404 on fetch.	Closest shipped timed-duel format; majors + leverage, no escrowed stake. Season model worth learning from for Ladder.	Verify details
Duel Duck (duelduck.com)	Substitute	Yes/no prediction “duels” collateralized in any SPL token; creator-set rules; admin or self resolution; smart-contract payout. Enabled custom predictions on any SPL token (March 2026).	Uses the word “duel” for prediction, not trading. Namespace overlap; betting-on-outcome vs competing-on-trades.	Verified

Name	Type	What it is (verified)	Relevance	Status
Kairos TradeX (Kairos Trade)	Adjacent	Trading terminal for Polymarket/Kalshi prediction markets; a16z crypto + Geneva Trading, \$2.5M seed Feb 2026; ran a trading championship with \$30K across Net PnL / ROI / Volume pools.	Tournament format on prediction markets; sponsored, many-player, not head-to-head.	Verified
Big Dance (bigdance.fun)	Substitute	Tournament prediction market: buy tokens on any March Madness team; loser pools merge into the winner's; 100% onchain; 50% referral fee share.	Event bracket, not trading skill. Referral-share model is a GTM idea.	Verified
Scorely	Not relevant	Crypto sportsbook on Solana with leverage odds trading; ~\$177K market cap token.	Sports betting. Exclude from table.	Verified
The Stonk Board (thestonkboard.com)	Ecosystem tool	Third-party dashboard: top 100 StonkFun reward coins bonded 3+ days, APR/APY, holder payouts. No games.	Confirms no StonkFun-native or third-party duel/competition product exists as of 2026-09-11. Possible allowlist data source.	Verified
Onboard (tryonboard.fun)	Adjacent, unverified	Listed as "Daily Solana Trading Challenge."	Possible daily-competition substitute.	Verify
PlayBabylon	Adjacent	Multiplayer prediction game for humans and AI agents; 490K trades across 1,171 agents in five days of launch (Solana roundup, March 2026).	Evidence that agent-vs-human arenas draw activity. Supports the Phase 2 agent note.	Verified via roundup
fomo (update)	Substitute, fast-follow	Series B \$75M in June 2026 after a \$17M Series A (late 2025) and \$2M angel (early 2025); ranked third in Solana 7-day protocol revenue (Solana Compass).	Sharpens CL-5: the fast-follow risk is a \$75M-funded company, not a startup.	Verified

Not found: any product running 1v1 duels with escrowed stakes on real spot fills of memecoins or StonkFun coins. DefiLlama raises search surfaced no funded “trading duel” company; the only funded adjacent names are fomo (social trading) and Kairos (prediction-market terminal).

Carry into Round 4: add TradeFighters (verify first), PerpClash, Duel Duck, Kairos, Big Dance, and the fomo funding update to the competitor table; note The Stonk Board as evidence the StonkFun niche is unoccupied; Colosseum Copilot search still outstanding (token required).

Round 4: Competitor landscape (2026-09-11)

Founder’s original text (unedited): the Draft v1 competitor table, resubmitted unchanged (SolDuel, PvPBattle, fomo competitions / FomoVille, DUELPVP / LootScape, TradeFlow, Pumpcade / Polymarket, pump.fun bounties / livestreams, CEX trading competitions, Melee Markets) with the request to include additional names from the sweep. Full table text is in docs/turbin3-loi-draft-v1.md, Part A.3.

Attack 1. Your own format now resembles your zero-traction competitor, and the table does not address it. SolDuel’s model is “pick a champion token, best % change wins.” A rival-coin duel where both players buy at the start and hold to the buzzer is the same game. The SolDuel row has to say why Rug Royale is different: trading stays open for the whole window so timing decides it, windows are short, the launch modes are unranked and freeroll, and the coin communities give it a rivalry SolDuel never had. It should also carry the other reading of SolDuel’s zero, that the category may have no demand. > **Founder:** “Rug Royale let’s you trade within the whole window so timing and skill play a part, and centered around coin communities, plus the transfer fees via StonkFun give innate skin in the game that’s missing from SolDuel.” > **AI counter:** Accepted. Wording fix: the transfer fee is community skin in the game (holders earn drip from duel volume); it is a cost to the duelists, so the format favors standard-mode coins for players. > **Decision:** Valid. SolDuel row rewritten with the differentiation and both readings of the zero.

Attack 2. Two direct competitors from the sweep are missing. TradeFighters (real-time duels between traders with a leaderboard; site is currently a logo) and PerpClash (live timed 1v1 duels on BTC/ETH/SOL perps with a season prize vault). > **Founder:** “Tradefighters, while describing a similar concept, has no active product. PerpClash runs a similar idea via perps, which we aren’t competing in. We may have to compete here for mindshare though.” > **AI counter:** “No active product” is one fetch on one day; verify via the TradeFighters X account before submission and write “pre-launch” if that is what it is. PerpClash framed as mindshare competition in a different market. > **Decision:** Valid. Both rows added; TradeFighters status marked “verify.”

Attack 3. The fomo row is out of date and understated. fomo closed a \$75M Series B in June 2026 after a \$17M Series A. “Could add 1v1; speed matters” describes a startup risk; the real row is that a \$75M company already running PnL competitions can ship 1v1 in a sprint. The 625k and 110M figures are unsourced. > **Founder:** “The original research I did here needs an update, they could definitely do a 1v1 trading comp, but the stonkfun community focus, plus the 1vN ideas we have planned down the line, give us enough differentiation that we can get far ahead before they move in. We need to source those figures though.” > **AI counter:** Partially accepted. Roadmap (1vN) is not a moat; a funded incumbent can build it in the same sprint.

Defensible differentiation: StonkFun community focus, first in an unoccupied niche, permissionless protocol shape any frontend can embed. Row states the fast-follow risk plainly. > **Decision:** Partially valid. fomo row updated with funding; differentiation limited to community, niche, and protocol shape; “get far ahead” claim not used; figures to be sourced.

Attack 4. Nothing in the table says the StonkFun niche is open. The sweep found no StonkFun-native or third-party duel or competition product; the only third-party tool is a reward-coin dashboard. The gap column never mentions rival coins, community rivalry, or drip. > **Founder:** “We need to update for our stonkfun niche pivot, which came out when we were thinking of an ideal customer.” > **Decision:** Valid. Niche-open statement added with The Stonk Board as evidence; gap column rewritten around the rival-coin community format.

Attack 5. Melee is labeled Partner. Round 2 relabeled it a prospective venue with no contact. pump.fun’s “\$20M/week creator payouts” is unsourced, and pump.fun launched its own social trading feature on 2026-08-07, so it belongs next to fomo as a fast-follow risk. > **Founder:** “Although I think of melee as a partner, if they ship live trading sessions that could be a competitor as well. We can mitigate this I think of just letting users get in on the action on our platform, which I think benefits both of us truthfully.” > **AI counter:** Partner-or-competitor is the right nuance. Warning: “get in on the action on our platform” means native spectator markets, which is the betting regulatory surface the design deliberately outsourced; native markets stay a v3 decision gated on counsel. The mitigation for Melee competing is that Melee needs content and duels are content. > **Decision:** Valid. Melee relabeled “prospective venue / possible competitor”; pump.fun row split into channel and fast-follow; native markets remain v3 gated on counsel.

Attack 6. The assignment requires a stated primary-research line (DefiLlama, Colosseum, GitHub, X) and a PMF adjustment from the competitor check. Neither appears. > **Founder:** “I searched DefiLlama, Colosseum, Github and X. The landscape helped me adjust my expectations to who could be a partner vs. competitor, and also risks if big platform wants to move into our area.” > **AI counter:** Recorded as stated. AI re-ran DefiLlama, GitHub, and X on 2026-09-11; the founder’s Colosseum result is not documented in this folder; founder to add findings or state nil. > **Decision:** Valid. Research line and PMF adjustment paragraph added to Section 3.

Attack 7. Smaller rows need updates from verified facts. PvPBattle verified as custodial, virtual positions vs Binance marks, 30s–15min battles, up to 25x, example pots 0.5–4 SOL, no published traction; “weaker legal posture” is opinion. Duel Duck, Kairos TradeX, and Big Dance are substitutes worth one row each. > **Founder:** “We will need to update this with verified facts and research.” > **Decision:** Valid. Rows updated from the sweep; opinion language removed.

Resulting changes carried forward: Section 3 assembled with 15 rows, research line, niche-open statement, and PMF adjustment.

Founder verification (2026-09-11, verbatim): “Tradefighters.fun has a product broadly similar to ours, but doesn’t have the specific market sharpening around the solana eco or stonkfun eco, plus the roadmap of our plans and granularity of bets. perpclash is a similar idea, but it doesn’t look live, and focuses only on perps.” → Rows updated; open items closed. Colosseum result: Pregame (sports-betting competitions) only.

Note on provenance: the competitor table submitted in this round was researched and iterated across the August 2026 deep dives (v1–v3) before the assignment; this round added the 2026-09-11 sweep and the pivot-driven rewrites.

Round 5: Founder-market fit (opened 2026-09-11)

Founder's original text (unedited): > I've been a data analyst since 2015, and learned advanced data analytics with python (completed springboard bootcamp) in 2021. I've been in crypto since 2014, built a successful mining business (BTC, LTC, and Dash) in 2017-2018, and a web3 consulting company in 2021-2024. I help my clients built, launch, and improve onchain applications. I have been coding since 2019, and buidling full stack dapps since 2021. I've built a successul miniapp on farcaster, which placed top 10 in the Encode hackathon, called 111words, which let's players build a daily writing habit and turn their posts into tradebale coins. I pivoted to solana 6 months ago, building a finalist product for the Bags Hackathon, solenrich. Solenrich is a x402 agentic intelligenece platform for agents trading on solana. I focus now on building agentic products on solana, x402, trading, and leverage all aspects of my background to build comprehensive and successful products

Attack 1. The one skill the capstone is graded on is not mentioned. The deliverable is an Anchor program with an escrow, an internal AMM, and settlement logic, tested in LiteSVM. The text says full-stack dapps since 2021, a Farcaster miniapp, and a Solana data API. None of those is a Solana program. Your Anchor and Rust level is the first thing the panel wants to know, and an honest "beginner, this cohort is where I learn it" is a better answer than silence.

Attack 2. There is no motivation. The assignment asks for background, skills, motivation, and network. This is a resume. It never says why trading duels, why StonkFun, or why you. The opening-list attack that the idea came from a scored shortlist rather than a personal itch still stands. If there is a moment, a call-out you could not settle or a stream where the argument happened, that is the paragraph.

Attack 3. Network and distribution assets are absent, and you have some. Distribution is the stated hard part of this product, and the section names no community you have a voice in. But the text contains assets it does not claim: the Bags hackathon put you in front of a Solana launchpad team and its finalist cohort, which is the closest thing to a StonkFun-shaped relationship you have. 111words gave you a Farcaster audience. Three years of web3 consulting gave you clients who launch onchain products. Name them as channels or the panel assumes you have none.

Attack 4. "Successful" appears twice with no number. Successful mining business, successful miniapp. Top 10 at Encode and finalist at Bags are real signals, but they need dates, links, and one metric each: users, revenue, posts created, duels of the mining era's scale. A panel discounts unquantified adjectives.

Attack 5. The credentials are not mapped to the risks in your own LOI. Your PMF section says the top unvalidated assumption is whether skill shows up at short windows, and the fix is a backtest. You are a data analyst with Python since 2015. That is the strongest founder-fit claim in the document and it is not made. Likewise, 111words turned posts into tradeable coins, which is the entertainmentFi thesis built once already, and SolEnrich's wallet graph is the integrity layer for ranked mode. Each credential should point at a specific risk it retires.

Attack 6. The closing focus statement points away from the capstone. “I focus now on building agentic products on Solana, x402, trading.” Rug Royale’s capstone is a human-versus-human game; agents are Phase 2. A panel will ask why an agent-focused builder is building a human game. Either connect it, agents enter the same duels through x402 in Phase 2 and SolEnrich is the data layer, or cut it.

Attack 7. Solo or team is not stated. The assignment says “your team’s background.” Say solo, or name the people from the team formation post and what each does.

Not attacks, for the final pass: “a a,” buidling, successul, intelligenece, tradebale, “let’s” should be “lets,” “help my clients built” should be “build.”

Founder rebuttals (2026-09-11, verbatim) and decisions:

A1: “I’ve been building in crypto since 2021. I have deep knowledge of blockchain primitives and onchain patterns, and am currently in Turbin3. I’m also skilled at building with AI, and can build within any gaps.” **AI counter:** Anchor level still not stated. The teammate named in A7 (Vineet / RustRogue, Rust and Solana program lead) answers most of the attack. One honest clause on the founder’s own Anchor level remains. **Decision:** Partially valid. Section states the program lead and the founder’s Anchor level in one clause each.

A2: “Building and trading onchain is something I’ve done for a long time, and I’m and early participant to pump.fun and fomo. I’ve watched entertainmentFi build from the ground up, and I have a key insight that the game theory mechanics of crypto also apply to our interactions themselves. I’ve also built solenrich.com, which brought me deep insight to trading bots and mechanics. Stonkfun I was an early participant in, and I understand yield and community mechanics both from my mining days, and when I was chief operating officer for a DAO in 2022.” **Decision:** Valid, and the rebuttal supplies the motivation paragraph: early participant in pump.fun, fomo, and StonkFun; the insight that crypto’s game-theory mechanics apply to trader interactions themselves; DAO COO in 2022 added to background.

A3: “I have distribution within stonkfun and bags communities. I also have a twitter account (@oxSardius, go look it up) with over 2300 followers across crypto, with specific heavy hitters in big solana protocols. I’m confident I can get this in front of the right people.” **Decision:** Valid. Network section names StonkFun and Bags communities, X @oxSardius with 2,300+ followers including people at major Solana protocols, the Turbin3 cohort, and the Farcaster audience from 111words.

A4: “Sure, I earned over \$200K in my mining business, and 111words won a top prize at encode. Solenrich has over 30 unique users and 700 transactions since inception.” **Decision:** Valid. Figures added; links to be attached by the founder.

A5: “you go ahead and map those, since I obviously have that in mind, but I feel this attack is just for the sake of it. Obviously it’s a good fit, and these map cleanly to build this product, I had the idea in the first place anyway with the combo of my background.” **AI counter:** The fit exists; the attack was that the document did not state it, and a panel reads the text. Downgraded. **Decision:** Partially valid. AI maps each credential to the LOI risk it retires; founder edits.

A6: “I don’t understand this one, I build agents onchain, and use agents to build onchain. Agents are mostly trading onchain, and I have deep experience interacting with and seeing what traders need. I also have been a lifelong gamer since 7 years old, and understand gamification and game theory in crypto deeply. I originally got into crypto in 2014 seeing virtual gold mined in WoW. So connect it for me.” **AI counter:** Connected: WoW gold (2014) → mining → DAO → 111words (writing into tradeable coins) → SolEnrich (data for trading agents) is one thesis, finance as a game people play. Rug Royale is the human arena; agents enter the same duels via x402 in Phase 2 with SolEnrich as the data layer. The through-line becomes the motivation paragraph. **Decision:** Valid as a wording problem, resolved by the connection above; the focus statement is rewritten rather than cut.

A7: “I plan on building with another builder I met in the program: Vineet, RustRogue. He is a highly skilled rust and solana coder who has built Conduit.” **Decision:** Valid. Team of two stated; roles assigned; Conduit link to be attached by the founder.

Resulting changes carried forward: FMF rewritten in first person with team, background with figures, motivation through-line, credential-to-risk map, and network. Founder to attach links (Encode result, Bags finalist, SolEnrich, 111words, Conduit) and correct the Anchor-level clause.

Round 6: Phase 2 design decisions and granularity check (2026-09-11)

Founder decisions (verbatim): “Explain 1 I don’t understand. 2 - Yes. 3. Yes. 4. Yes. 5. Yes. 6. yes 7. yes. 8. yes, good. 9. yes, keep it optional.” After explanation of Decision 1 (per-player coin with shared quote covers both “same asset” and “pick from a list”): “good we settled it then.”

#	Decision	Founder	Applied as
1	Per-player coin, shared quote asset	Yes (after explanation)	Duel.coin_a, Duel.coin_b, Duel.quote_mint; one or two pools per duel
2	Simulated independent liquidation at settlement	Yes	settle values each player against own pool reserves without mutation

#	Decision	Founder	Applied as
3	Token-2022 interface for vaults and pools	Yes	<code>token_interface</code> , <code>transfer_checked</code> ; drip counted if received before settle (mainnet)
4	Init split between <code>create_duel</code> and <code>join_duel</code>	Yes	pools and creator vaults at create; opponent vaults (and pool B if new) at join
5	<code>sponsor_prize</code>	Yes	UC-3; one sponsor per duel; returns on cancel/refund; split on tie
6	Permissionless <code>emergency_refund</code>	Yes	UC-8; admin appears only in <code>init_config</code>
7	Duel account is the oracle	Yes	<code>close_duel</code> never closes Duel; event is a notification
8	Tunables in <code>Config</code>	Yes	tiers, windows, <code>pool_seed_ratio</code> , <code>allowed_mints</code> , <code>settler_tip</code> , <code>quote_mint</code>
9	Settler tip	Yes, optional	<code>settler_tip_lamports</code> , 0 disables

Granularity check against the four rules: all four pass on the final nine-handler set; the table with reasoning is in the refined LOI, Phase 2. Changes since Draft v1: `refund` split (GC-3), init moved (GC-1), `sponsor_prize` added (GC-5), oracle claim corrected (GC-6), play-money bankroll and PDA mint authority stated (GC-7, GC-13), admin trust removed (GC-10), single-signer rule for cancel (GC-11), tunables on-chain (GC-16). Deferred to Phase 6: on-chain host share (GC-8), on-chain allowlist (GC-15), mainnet indexer (GC-9).

Sources checked for this log (2026-09-09)

- perpclub.com (site root returned 404 on fetch; description from search snippet: “1v1 perpetual trading duels on live Solana markets ... BTC, ETH, and SOL ... no deposit required, up to 20x leverage ... season points feed the prize vault ... top 100 traders split the pool”). Verify directly before the final.
- pvpbattle.xyz (fetched): “Positions are virtual and settle against the live Binance mark price, so nothing is executed on an exchange”; custodial deposit/withdraw; 30 seconds to 15 minutes; up to 25x; example pots 0.5–4.0 SOL; no traction figures published.
- cryptobriefing.com, “Fomo and Pump.fun battle for trader flow” (fetched): fomo weekly revenue crossed \$2M; fomo briefly overtook Axiom in daily fees on 2026-08-06; pump.fun social trading feature launched 2026-08-07; pump.fun reported new ATH daily active users after launch.
- solanacompass.com, Colosseum Frontier Hackathon 2026 winners: 26 winners, grand champion CrowdBrain (DePIN); no trading-duel project in the winners list on first pass.

- Search results also surfaced sol-pvp.com and SOL Arena (blockchaingame.world) as Solana PvP games; not yet verified.